Survival Analysis

Lecture notes, Lectures 1–10

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Lecture 1. Introduction

1 Course Overview

- Textbook: Klein & Moeschberger, *Survival Analysis*, 2nd ed. (K&M), Chapters 2–5, 7–9, 11, 12.
- Methodology and application, with part of the theory.
- Prerequisites: basic probability theory, statistical inference, and R programming.

2 What Is Survival Data?

Definition 1.1: Survival data

Survival data track the *time until an event of interest occurs*:

$$\text{Exposure (clock starts)} \xrightarrow{\text{survival time}} \text{Event (clock ends)}.$$

Examples: diagnosis to death; treatment to recurrence; machine failure; promotion; divorce; unsubscription.

Example 1.2: Worcester Heart Attack Study (WHAS100)

Survival after hospitalisation for a heart attack (MI):

ID	Female	Age (years)	Follow-up (years)	Event (death)
16	0	39	7.2	0
66	1	91	1.0	1
71	0	65	5.1	0
55	0	80	5.3	0
61	0	90	0.3	1

Question: what is the mean survival time after hospitalisation?

- *Wrong approach 1* (known survival times only): $(1.0+0.3)/2 = 0.65$, an **underestimate** (longer survivors ignored).
- *Wrong approach 2* (follow-up = survival): $(7.2+1.0+5.1+5.3+0.3)/5 = 3.78$, also an **underestimate** (IDs 16, 71, 55 survived longer).

Acknowledging censoring. 7.2^+, 1.0, 5.1^+, 5.3^+, 0.3: ID 66 survived exactly 1 year; ID 16 *at least* 7.2 years. The data carry **incomplete (partial) information**.

Remark 1.3: Why both naive estimates are biased downward

With true time T_i and follow-up C_i we observe $Y_i = \min(T_i, C_i) \le T_i$, so $\mathrm{E}(\bar{Y}) \le \mathrm{E}(T)$ (approach 2). Approach 1 conditions on $\{T_i \le C_i\}$, preferentially discarding long survivors (Exercise 1.1).

3 What Is Special About Survival Data?

- Classical statistics assumes i.i.d. data $T_1, \ldots, T_n$; survival data are **incomplete**.
- A survival problem needs:
  (1) the **event of interest**;
  (2) the **time origin**;
  (3) the **time scale**.
- Incompleteness: **censoring** or **truncation**: right censoring (RC), left censoring (LC), interval censoring (IC), left truncation (LT), right truncation (RT).

4 Types of Censoring

4.1 Right censoring

Definition 1.4: Right censoring (RC)

The event happens *after* a known time point: $T_i > C_i$.

Example 1.5: Time to death in a breast cancer trial

T_i = time from randomization to death. If the study ends 9 months after treatment with the patient alive, $T_i > 9$.

4.2 Left censoring

Definition 1.6: Left censoring (LC)

The event happened *before* a known time point: $T_i < C_i$.

Example 1.7: Time to first use of marijuana

"When did you first use marijuana?"

- the exact age (uncensored);
- "never used it" (right censored at the current age);
- "used it but cannot recall when" (left censored at the current age).

Both left and right censoring: **double censoring**.

4.3 Interval censoring

Definition 1.8: Interval censoring (IC)

The event happened within an interval: $T_i \in (L_i, R_i]$.

Example 1.9: Time to tumour recurrence

MRI every 5 months after surgery; tumour absent at month 10, detected at 15: $10 < T_i \leq 15$.

Remark 1.10

RC at C is $T \in (C, \infty)$, LC at C is $T \in [0, C)$: special cases of IC.

5 Types of Truncation

5.1 Left truncation

Definition 1.11: Left truncation (LT)

Only individuals surviving long enough are **included**: with entry time V_i, T_i is observed only if $T_i \geq V_i$ (the conditional law given $T_i \geq V_i$).

Example 1.12: Longevity of elderly residents of a retirement community

Only those surviving to entry age enter the centre. Ignoring LT gives **length-biased sampling**: long lifetimes over-represented.

5.2 Right truncation

Definition 1.13: Right truncation (RT)

Only individuals who *experienced the event* by a given time are included: T_i observed only if $T_i < R_i$.

Example 1.14: Age at onset of diabetes

Patients report their age of onset T_i retrospectively; with R_i the age at study, T_i is observed given $T_i < R_i$.

Example 1.15: HIV incubation time (Lagakos et al., 1988)

258 adults and 37 children infected with HIV who developed AIDS by June 30, 1986. With X = age at infection and R = age at AIDS, $R - X$ is known only for those infected *and* diagnosed within the window: right truncation.

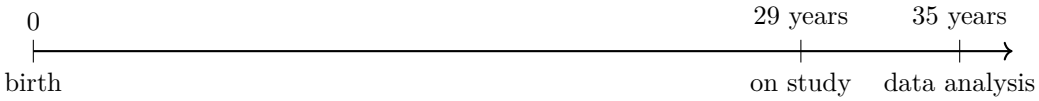
5.3 Censoring vs. truncation

Censoring	Subject *detected*; value known only to be below and/or above some number.
Truncation	Subject detected *only if* its value exceeds (or is below) some number; then fully known.

A truncated subject never enters the data set.

6 CVD Example: Time Scales, Truncation and Censoring Together

A study enrols CVD-free subjects; the event is CVD. A subject enters at age 29; data are analyzed at 35.



Two time scales:

- origin = birth: T_i^1 = age at CVD;
- origin = study entry: T_i^2 = years on study until CVD.

Both may be right censored; T_i^1 is usually of more interest.

Scenario	T_i^1 (age)	T_i^2 (time on study)
CVD at age 34	$T_i^1 = 34$ observed *given* $T_i^1 \geq 29$: **LT at 29**	$T_i^2 = 34 - 29 = 5$: no truncation, no censoring
No CVD by 35	$T_i^1 > 35$ given $T_i^1 \geq 29$: **LT at 29 and RC at 35**	$T_i^2 > 6$: **RC**, not LT
Enrolled at 30, already had CVD before 30	$T_i^1 < 30$: **LC at 30**	—

7 Definitions and Censoring Patterns

- **Failure time** (survival time, event time): the length of "time" to an event.
- **Time origin**: objective and meaningful; subjects comparable at the origin.
- **Time scale**: usually calendar time; others possible, e.g. QALY.

Definition 1.16: Equal or unequal censoring times

- **Type I**: common entry time and equal censoring times.
- **Generalized Type I**: staggered entry, censoring times known at entry.
- **Random**: losses to follow-up (other deaths, relocation, withdrawal); censoring times known only for the censored.

Record the actual censoring *times*, not just the censoring indicator.

8 Possible Simple (Naive) Methods

1. **Dichotomise at** t_0 and use binary methods (chi-square, logistic regression).
   Drawbacks: censoring before t_0; one point of the curve only; arbitrary t_0.

2. **Log-linear model.** Drawbacks: dependence on short times; log-normality may fail; no censoring.

Both can serve as a prelude.

9 Summary

- RC, LC, IC, LT, RT (plus double censoring); LT and RC are most common.
- **Censoring**: T_i known only to be $>$ or $<$ a value, or in an interval.
- **Truncation**: T_i observed conditionally on $T_i >$ or $<$ a value.
- Mixtures are more complex.

10 Selected exercises

Motivation. Proofs of the biases: discarding censored subjects and right truncation shorten, left truncation lengthens observed lifetimes; the origin matters except for the exponential.

Exercise 1.1 (Approach 1 is biased downward)

Let $T \geq 0$ with $\mathrm{E}(T) < \infty$ and let C be independent of T with $P(T \leq C) > 0$.
(a) Prove Chebyshev's association inequality: if φ is nondecreasing, g is nonincreasing, both are bounded or $\varphi(T), g(T), \varphi(T)g(T)$ are integrable, then $\mathrm{E}[\varphi(T)g(T)] \leq \mathrm{E}[\varphi(T)]\,\mathrm{E}[g(T)]$.
(b) Show that $\mathrm{E}(T \mid T \leq C) \leq \mathrm{E}(T)$.

Solution [Supplement].

(a) With T' an independent copy, $\varphi(s) - \varphi(t)$ and $g(s) - g(t)$ have opposite signs, so

$$\{\varphi(T) - \varphi(T')\}\{g(T) - g(T')\} \leq 0.$$

and taking expectations (T, T' i.i.d.)

$$2\mathrm{E}[\varphi(T)g(T)] - 2\mathrm{E}[\varphi(T)]\mathrm{E}[g(T)] \leq 0.$$

(b)

Step 1. $g(t) = P(C \geq t) \in [0, 1]$ is nonincreasing; by independence and Fubini, $P(T \leq C) = \mathrm{E}[g(T)]$, $\mathrm{E}[T\, I(T \leq C)] = \mathrm{E}[T\, g(T)]$.

Step 2. (a) with $\varphi(t) = t$: $\mathrm{E}[Tg(T)] \leq \mathrm{E}(T)\mathrm{E}[g(T)]$.

Step 3. Divide by $\mathrm{E}[g(T)] = P(T \leq C) > 0$:

$$\mathrm{E}(T \mid T \leq C) = \frac{\mathrm{E}[T\, I(T \leq C)]}{P(T \leq C)} \leq \mathrm{E}(T).$$

Approach 1 estimates the left side: biased downward for any censoring law. ■

Exercise 1.2 (Truncation shifts the observed distribution)

Let $T \geq 0$ have survival function S.
(a) Let V be an entry time independent of T with $P(T \geq V) > 0$. Show that $P(T > t \mid T \geq V) \geq S(t)$ for every t, i.e. left-truncated lifetimes are stochastically larger than T.
(b) Let R be independent of T with $P(T < R) > 0$. Show that $P(T > t \mid T < R) \leq S(t)$ for every t.

Solution [Supplement].

(a) With $g(s) = P(V \leq s)$ nondecreasing, $P(T \geq V) = \mathrm{E}[g(T)]$ and $P(T > t, T \geq V) = \mathrm{E}[I(T > t)g(T)]$. Exercise 1.1(a) with $\varphi = I(\cdot > t)$ and the nonincreasing $-g$ gives

$$\mathrm{E}[I(T > t)g(T)] \geq P(T > t)\,\mathrm{E}[g(T)].$$

Divide by $\mathrm{E}[g(T)] > 0$.

(b) $g(s) = P(R > s)$ is nonincreasing, $P(T < R) = \mathrm{E}[g(T)]$, $P(T > t, T < R) = \mathrm{E}[I(T > t)g(T)] \leq P(T > t)\mathrm{E}[g(T)]$ by Exercise 1.1(a); divide.

This is length-biased sampling (retirement community) and the opposite bias (diabetes). ■

Exercise 1.3 (Changing the time origin)

Let T^1 (age at the event) be continuous with survival S_1 and hazard h_1, and let $a \geq 0$ with $S_1(a) > 0$. A subject enters the study at age a and is followed on the study scale $T^2 = T^1 - a$, conditionally on $T^1 \geq a$.
(a) Show that, given $T^1 \geq a$, T^2 has survival $S_1(a+u)/S_1(a)$ and hazard $h_1(a+u)$, $u \geq 0$.
(b) Assume $S_1 > 0$ on $[0, \infty)$. Show that the conditional law of T^2 equals the law of T^1 for every $a \geq 0$ if and only if T^1 is exponential.

Solution [Supplement].

(a) $P(T^2 > u \mid T^1 \geq a) = S_1(a+u)/S_1(a)$ (continuity), and $-\frac{\mathrm{d}}{\mathrm{d}u} \log S_1(a+u) = h_1(a+u)$.

(b)

Step 1. If $T^1 \sim \mathrm{Exp}(\lambda)$, then $S_1(a+u)/S_1(a) = e^{-\lambda u} = S_1(u)$.

Step 2. Conversely, $S_1(a+u) = S_1(a)S_1(u)$ makes $\psi = -\log S_1$ additive, $\psi \geq 0$ nondecreasing.

Step 3. $\psi(q) = q\,\psi(1)$ for rational $q \geq 0$; for rationals $q_1 \leq x \leq q_2$, $q_1\psi(1) \leq \psi(x) \leq q_2\psi(1)$, so $\psi(x) = \lambda x$, $\lambda = \psi(1) > 0$ (as $S_1(\infty) = 0$).

So outside the exponential model, the age scale (with LT) and the study scale (without) answer different questions. ■

Lecture 2. Basic Quantities

1 Review: Classifying Incomplete Data

Situation	Type
Trial from randomization; no event by month 24 at study end.	RC
Tumour present at baseline; onset only known to precede it.	LC
Annual screening; disease-free at year 2, first positive at year 3.	IC
Registry enrols only patients alive at clinic entry (up to 3 years after diagnosis); followed to death or study end.	LT + (RC or event)
Cohort built in 2020 from obituaries for 2010–2019; age at death only.	RT

2 Survival Function

Let $X \geq 0$ be a random failure time.

Definition 2.1: Survival function

For continuous X with pdf f and cdf $F(x) = P(X \leq x)$,

$$S(x) = P(X > x) = 1 - F(x) = \int_x^\infty f(s)\,\mathrm{d}s.$$

the probability of surviving beyond x; non-increasing, $S(0) = 1$, $S(\infty) = 0$.

$S(x)$ is the area under the density right of x. (E.g. 1989 US survival curves stay near 1 until middle age, then drop steeply.)

3 Hazard Function

Definition 2.2: Hazard function

$$h(x) = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x \mid X \geq x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x)/P(X \geq x)}{\Delta x} = \frac{f(x)}{S(x)}.$$

- $f(x)\,\mathrm{d}x \approx S(x) - S(x + \mathrm{d}x)$: failure probability at x, *unconditionally*.
- $h(x)\,\mathrm{d}x$: failure probability just after x, *given* alive at x.

Remark 2.3: h is a rate, not a probability

$h(x)$ can exceed 1; only $h(x)\Delta x$ approximates a conditional probability.

3.1 Shapes of hazard functions

(i) **Constant**: exponential.

(ii) **Increasing**: ageing industrial systems and cars.

(iii) **Decreasing**: e.g. transplants (risk highest after the operation).

(iv) **Bathtub**: population mortality (infancy high, mid-life low, old age rising).

(v) **Hump-shaped**: e.g. breast cancer treatment, where high-risk patients die early.

3.2 Examples

Example 2.4: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$ and

$$S(x) = \int_x^\infty \lambda e^{-\lambda s}\,\mathrm{d}s = \big[-e^{-\lambda s}\big]_x^\infty = e^{-\lambda x}, \qquad h(x) = \frac{f(x)}{S(x)} = \lambda.$$

Constant hazard = memorylessness: $P(X > x + t \mid X > x) = e^{-\lambda t}$.

Example 2.5: Weibull(α, λ)

$S(x) = \exp(-\lambda x^\alpha)$. Then

$$f(x) = -\frac{\mathrm{d}S(x)}{\mathrm{d}x} = \alpha\lambda x^{\alpha-1}\exp(-\lambda x^\alpha), \qquad h(x) = \frac{f(x)}{S(x)} = \alpha\lambda x^{\alpha-1}.$$

Increasing for $\alpha > 1$, constant for $\alpha = 1$, decreasing for $\alpha < 1$.

4 Cumulative Hazard and the S–h Relationship

Motivation. S, f, h, H describe the same law; $S = \exp\{-H\}$ moves between them.

Theorem 2.6: Continuous case

Define the **cumulative hazard** $H(x) = \int_0^x h(s)\,\mathrm{d}s$. If X is absolutely continuous,

$$H(x) = -\ln S(x), \qquad S(x) = \exp\{-H(x)\} = \exp\Big\{-\int_0^x h(s)\,\mathrm{d}s\Big\}.$$

Proof [Supplement]. Since $f(s) = -S'(s)$,

$$H(x) = \int_0^x \frac{f(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \frac{S'(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \mathrm{d}\{\ln S(s)\} = -\big[\ln S(x) - \ln S(0)\big] = -\ln S(x),$$

using $S(0) = 1$. ■

Summary of relations (continuous X)	
$f(t) = \dfrac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t)$	$F(t) = 1 - S(t)$
$h(t) = \dfrac{f(t)}{S(t)} = -\dfrac{\mathrm{d}}{\mathrm{d}t}\ln S(t)$	$H(t) = \displaystyle\int_0^t h(s)\,\mathrm{d}s = -\ln S(t)$
$S(t) = \exp\{-H(t)\}$	$f(t) = h(t)\exp\{-H(t)\}$

Any one of f, F, S, h, H determines the others.

5 Discrete Failure Times

At $0 = x_0 < x_1 < \cdots < x_d$ let $P(X = x_j) = p_j > 0$, $\sum p_j = 1$ (rounding or grouping). Then

$$S(x) = P(X > x) = \sum_{j=1}^{d} P(X = x_j)\, I(x_j > x),$$

a right-continuous step function.

Definition 2.7: Discrete hazard

$$h(x_j) = P(X = x_j \mid X \geq x_j) = \frac{p_j}{S(x_{j-1})} = \frac{p_j}{\sum_{k=j}^{d} p_k}, \qquad j = 1, \ldots, d.$$

Here $h(x_j)$ *is* a conditional probability: $0 < h(x_j) \leq 1$.

Since $p_j = S(x_{j-1}) - S(x_j)$,

$$h(x_j) = \frac{S(x_{j-1}) - S(x_j)}{S(x_{j-1})} = 1 - \frac{S(x_j)}{S(x_{j-1})}.$$

Example 2.8: $p_j = 1/3$, $j = 1, 2, 3$

$$S(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 2/3, & 1 \leq x < 2 \\ 1/3, & 2 \leq x < 3 \\ 0, & x \geq 3 \end{cases} \qquad h(1) = \frac{1/3}{1} = \frac{1}{3}, \quad h(2) = \frac{1/3}{2/3} = \frac{1}{2}, \quad h(3) = \frac{1/3}{1/3} = 1.$$

The last hazard is 1: survivors to the last support point fail there.

Theorem 2.9: Product-limit representation

$$S(x_j) = \frac{S(x_j)}{S(x_{j-1})} \frac{S(x_{j-1})}{S(x_{j-2})} \cdots \frac{S(x_1)}{S(x_0)} S(x_0) = \prod_{i: x_i \leq x_j} \frac{S(x_i)}{S(x_{i-1})} = \prod_{i: x_i \leq x_j} [1 - h(x_i)],$$

(telescoping, $S(x_0) = 1$).

This is the population Kaplan–Meier form (Lecture 5).

Remark 2.10: Two definitions of the discrete cumulative hazard

K&M define $H(x_j) = \sum_{i: x_i \leq x_j} h(x_i)$, for which $S = e^{-H}$ fails. Cox & Oakes (1984) define

$$H(x_j) = -\sum_{i: x_i \leq x_j} \log\left[1 - h(x_i)\right] \approx \sum_{i: x_i \leq x_j} h(x_i),$$

which restores $S = e^{-H}$ exactly ($-\log(1-u) \approx u$ for small u).

6 Mean and Mean Residual Life

Motivation. The mean residual life is the expected remaining lifetime of a survivor, computable from S.

Theorem 2.11: Mean as area under S

For $X \geq 0$, $\mathrm{E}(X) = \int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty S(t)\,\mathrm{d}t$.

Proof [Supplement]. Write $t = \int_0^t \mathrm{d}u$ and use Tonelli:

$$\int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty \int_0^t f(t)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_0^\infty S(u)\,\mathrm{d}u.$$

■

Definition 2.12: Mean residual life

$\mathrm{mrl}(x) = \mathrm{E}(X - x \mid X > x)$. For continuous X,

$$\mathrm{mrl}(x) = \frac{\int_x^\infty (t-x) f(t)\,\mathrm{d}t}{S(x)} = \frac{\int_x^\infty S(t)\,\mathrm{d}t}{S(x)},$$

the area under S right of x divided by $S(x)$; $\mathrm{mrl}(0) = \mathrm{E}(X)$.

Proof of the second equality [Supplement]. With $t - x = \int_x^t \mathrm{d}u$ and Fubini, $\int_x^\infty (t-x) f(t)\,\mathrm{d}t = \int_x^\infty \int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_x^\infty S(u)\,\mathrm{d}u$. ■

7 Quantiles

Definition 2.13: p-th quantile

The p-th quantile ($100p$-th percentile) is

$$x_p = \inf\{t : S(t) \leq 1 - p\}.$$

For continuous strictly decreasing S, $S(x_p) = 1 - p$; the median is $x_{0.5}$.

Example 2.14: Exponential(λ) — mean, mrl, median

$S(x) = e^{-\lambda x}$.

- $\mathrm{E}(X) = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = 1/\lambda$.
- $\mathrm{mrl}(x) = \dfrac{e^{-\lambda x}/\lambda}{e^{-\lambda x}} = \dfrac{1}{\lambda}$ (memorylessness).
- Median: $e^{-\lambda x_{0.5}} = 1/2 \Rightarrow x_{0.5} = (\ln 2)/\lambda$.

Example 2.15: Weibull quantile

$F(x_p) = p$ gives

$$1 - e^{-\lambda x_p^\alpha} = p \implies e^{-\lambda x_p^\alpha} = 1 - p \implies \lambda x_p^\alpha = -\ln(1-p) \implies x_p = \Big[\frac{-\ln(1-p)}{\lambda}\Big]^{1/\alpha}.$$

Median: $(\ln 2/\lambda)^{1/\alpha}$.

Example 2.16: LogNormal quantile

$Y = \ln X \sim N(\mu, \sigma^2)$, and ln is increasing:

$$p = P(X \le x_p) = P(Y \le \ln x_p) = \Phi\Big(\frac{\ln x_p - \mu}{\sigma}\Big) \implies x_p = \exp\left(\mu + \sigma\Phi^{-1}(p)\right),$$

so median$(X) = e^\mu$.

8 Median Residual Life

Definition 2.17: Median residual life

Preferred to the mrl for skewed distributions:

$$\text{mdrl}(x) = \text{median}(X - x \mid X > x),$$

equivalently

$$P(X - x > \text{mdrl}(x) \mid X > x) = \frac{1}{2} \iff \frac{P(X > x + \text{mdrl}(x),\, X > x)}{P(X > x)} = \frac{1}{2}.$$

For continuous X the numerator is $S(x + \text{mdrl}(x))$:

$$S(x + \text{mdrl}(x)) = \tfrac{1}{2}S(x) \quad \implies \quad \text{mdrl}(x) = S^{-1}\Big(\frac{S(x)}{2}\Big) - x.$$

Example 2.18: Exponential(λ)

$S^{-1}(y) = -(1/\lambda)\ln y$, so

$$\text{mdrl}(x) = -\frac{1}{\lambda}\ln\Big(\frac{e^{-\lambda x}}{2}\Big) - x = -\frac{1}{\lambda}(-\lambda x - \ln 2) - x = \frac{\ln 2}{\lambda}.$$

constant in x.

9 Weibull Mean Residual Life

Let $S(t) = \exp(-\lambda t^\alpha)$, $\lambda, \alpha > 0$, and $m(t) = \text{E}[T - t \mid T > t]$.

1. Derive $m(t)$. 2. Show $m(t)$ is constant when $\alpha = 1$. 3. For $\alpha > 1$, show $m(t)$ is decreasing and interpret.

Derivation: substitution $v = \lambda u^\alpha$

With $f(u) = \lambda\alpha u^{\alpha-1}e^{-\lambda u^\alpha}$, the numerator is

$$N(t) = \int_t^\infty (u-t)\,\lambda\alpha u^{\alpha-1}e^{-\lambda u^\alpha}\,\mathrm{d}u.$$

With $v = \lambda u^\alpha$, $\mathrm{d}v = \lambda\alpha u^{\alpha-1}\,\mathrm{d}u$, $u = (v/\lambda)^{1/\alpha}$, limits λt^α to ∞:

$$\begin{aligned} N(t) &= \int_{\lambda t^\alpha}^\infty \Big(\big(\tfrac{v}{\lambda}\big)^{1/\alpha} - t\Big)e^{-v}\,\mathrm{d}v = \lambda^{-1/\alpha}\int_{\lambda t^\alpha}^\infty v^{1/\alpha}e^{-v}\,\mathrm{d}v - t\int_{\lambda t^\alpha}^\infty e^{-v}\,\mathrm{d}v \\ &= \lambda^{-1/\alpha}\,\Gamma\big(1+\tfrac{1}{\alpha},\lambda t^\alpha\big) - t\,e^{-\lambda t^\alpha}, \end{aligned}$$

with $\Gamma(s,x) = \int_x^\infty v^{s-1}e^{-v}\,\mathrm{d}v$ the upper incomplete gamma function. Dividing by $S(t) = e^{-\lambda t^\alpha}$:

$$m(t) = \frac{N(t)}{S(t)} = \frac{1}{S(t)}\cdot\frac{1}{\lambda^{1/\alpha}}\,\Gamma\big(1+\tfrac{1}{\alpha},\lambda t^\alpha\big) - t.$$

- $\alpha = 1$: $\Gamma(2,\lambda t) = (1+\lambda t)e^{-\lambda t}$, so $m(t) = \frac{(1+\lambda t)}{\lambda} - t = \frac{1}{\lambda}$ (constant).
- $\alpha > 1$: $h(t) = \lambda\alpha t^{\alpha-1}$ increases, so $m(t)$ decreases (§10): the system *ages*.

10 mrl and Hazard Monotonicity

Motivation. Increasing risk means shrinking expected remaining life.

Theorem 2.19

Let $T \geq 0$ have survival S, density f, hazard $h = f/S$, and $m(t) = \mathrm{E}[T - t \mid T > t]$. Then

1. $m'(t) = h(t)m(t) - 1$;
2. if h is increasing then m is decreasing; if h is decreasing then m is increasing.

(1) Differential equation [Supplement]. $m(t) = \int_t^\infty S(u)\,\mathrm{d}u/S(t)$; by the quotient rule,

$$m'(t) = \frac{-S(t)\cdot S(t) - \big(\int_t^\infty S(u)\,\mathrm{d}u\big)S'(t)}{S(t)^2}.$$

With $S'(t) = -h(t)S(t)$,

$$m'(t) = \frac{-S(t)^2 + \big(\int_t^\infty S(u)\,\mathrm{d}u\big)h(t)S(t)}{S(t)^2} = -1 + h(t)\frac{\int_t^\infty S(u)\,\mathrm{d}u}{S(t)} = -1 + h(t)m(t).$$

■

(2) Monotonicity [Supplement]. By (1), $m'(t) \leq 0 \iff h(t)m(t) \leq 1$. Substituting $u \mapsto t+u$,

$$m(t) = \int_0^\infty \frac{S(t+u)}{S(t)}\,\mathrm{d}u = \int_0^\infty \exp\Big(-\int_t^{t+u} h(s)\,\mathrm{d}s\Big)\,\mathrm{d}u.$$

- h increasing: $\int_t^{t+u} h(s)\,\mathrm{d}s \geq u\,h(t)$, so

$$m(t) \leq \int_0^\infty e^{-uh(t)}\,\mathrm{d}u = \frac{1}{h(t)} \;\Rightarrow\; h(t)m(t) \leq 1 \;\Rightarrow\; m'(t) \leq 0.$$

- h decreasing: $\int_t^{t+u} h \le u\,h(t)$, so $m(t) \ge 1/h(t)$ and $m'(t) \ge 0$.

■

Example 2.20: Application to the Weibull

$h(t) = \lambda\alpha t^{\alpha-1}$.

- $\alpha < 1$: h decreasing $\Rightarrow$ m increasing.
- $\alpha = 1$: $m'(t) = \lambda m(t) - 1$, bounded solution $m \equiv 1/\lambda$.
- $\alpha > 1$: h increasing $\Rightarrow$ m decreasing.

11 Selected exercises

Motivation. K&M Section 2.8: a discrete constant hazard, a mixture with falling hazard, and a distribution recovered from its mrl.

Exercise 2.1 (K&M 2.13: geometric distribution)

Let $P(X = x) = p(1-p)^{x-1}$, $x = 1, 2, \ldots$, with $0 < p < 1$.
(a) Find the survival function of X.
(b) Find the discrete hazard of X and compare it with the exponential hazard.

Solution.

(a) For integer $k \ge 0$,

$$P(X > k) = \sum_{j=k+1}^{\infty} p(1-p)^{j-1} = p(1-p)^k \sum_{i=0}^{\infty} (1-p)^i = (1-p)^k.$$

So $S(x) = (1-p)^{\lfloor x \rfloor}$, $x \ge 0$.

(b) For $j \ge 1$,

$$h(j) = \frac{P(X = j)}{P(X \ge j)} = \frac{p(1-p)^{j-1}}{S(j-1)} = \frac{p(1-p)^{j-1}}{(1-p)^{j-1}} = p,$$

constant, as for the exponential ($P(X > j + k \mid X > j) = (1-p)^k$). At integers $S(j) = e^{-\lambda j}$ with $\lambda = -\log(1-p) > p$: the Cox–Oakes cumulative hazard $j\lambda$, whereas K&M gives jp and $e^{-jp} \ne S(j)$. ■

Exercise 2.2 (K&M 2.14: gamma mixture of exponentials)

Given $\Theta = \theta$, the lifetime X is exponential with hazard θ; the individual hazard Θ has the gamma density $\lambda^\beta \theta^{\beta-1} e^{-\lambda\theta}/\Gamma(\beta)$, $\theta > 0$, with $\lambda, \beta > 0$.
(a) Find the survival function of X.
(b) Find the hazard rate of X and describe its shape.

Solution.

(a) $S(x) = \mathrm{E}[P(X > x \mid \Theta)] = \mathrm{E}[e^{-\Theta x}]$, and

$$\mathrm{E}[e^{-\Theta x}] = \int_0^\infty \frac{\lambda^\beta \theta^{\beta-1} e^{-(\lambda+x)\theta}}{\Gamma(\beta)}\,\mathrm{d}\theta = \frac{\lambda^\beta}{(\lambda+x)^\beta} \int_0^\infty \frac{(\lambda+x)^\beta \theta^{\beta-1} e^{-(\lambda+x)\theta}}{\Gamma(\beta)}\,\mathrm{d}\theta = \Big(\frac{\lambda}{\lambda+x}\Big)^\beta,$$

(the last integrand is the Gamma$(\lambda + x, \beta)$ density).

(b) $H(x) = \beta\{\log(\lambda + x) - \log \lambda\}$, so

$$h(x) = H'(x) = \frac{\beta}{\lambda + x},$$

strictly decreasing from $\beta/\lambda = \mathrm{E}(\Theta)$ to 0: individual hazards are constant, but the frail (large Θ) fail first. Indeed $\Theta \mid X > x$ has density $\propto e^{-\theta x}\theta^{\beta-1}e^{-\lambda\theta}$, Gamma$(\lambda + x, \beta)$, and $h(x) = \mathrm{E}(\Theta \mid X > x)$. (The marginal is Lomax.) ■

Exercise 2.3 (K&M 2.17: distribution from the mean residual life)

A continuous lifetime X has mean residual life $\mathrm{mrl}(x) = x + 10$ for all $x \geq 0$.
(a) Find $\mathrm{E}(X)$.
(b) Find $h(x)$.
(c) Find $S(x)$.

Solution.

(a) $\mathrm{E}(X) = \mathrm{mrl}(0) = 10$.

(b)

Step 1. With $R(x) = \int_x^\infty S(u)\,\mathrm{d}u$, the mrl needs $S > 0$, and $R(x) = (x + 10)S(x)$, $x \geq 0$.

Step 2. R is continuous, so is $S = R/(x + 10)$; then $R' = -S$ exists and S is differentiable.

Step 3. Differentiating, $-S(x) = S(x) + (x + 10)S'(x)$, so

$$h(x) = -\frac{S'(x)}{S(x)} = \frac{2}{x + 10}.$$

(Or by §10: $1 = m'(x) = h(x)m(x) - 1$.)

(c) $H(x) = \int_0^x \frac{2}{u+10}\,\mathrm{d}u = 2\log\frac{x+10}{10}$, so

$$S(x) = e^{-H(x)} = \Big(\frac{10}{x + 10}\Big)^2, \qquad x \geq 0.$$

Check: $\int_x^\infty 100(u + 10)^{-2}\,\mathrm{d}u = (x + 10)S(x)$. Increasing mrl, decreasing hazard (§10). ■

Lecture 3. Basic Parametric Distributions and Models

1 Outline

- Parametric families: exponential, Weibull, extreme value, log-normal, log-logistic, gamma.
- Regression: accelerated failure time (log-linear), multiplicative (PH), additive hazards.

2 Exponential Distribution

Definition 3.1: Exponential(λ)

$f(x) = \lambda e^{-\lambda x}$, $S(x) = e^{-\lambda x}$, $h(x) = \lambda$ (constant), $H(x) = \lambda x$.

- Memoryless: the failure rate given survival to t does not depend on t.
- **Empirical check**: $\hat{H}(x)$ against x should be a line through the origin.
- Simple but too restrictive in practice.
- Weibull with $\alpha = 1$; gamma with $\beta = 1$.

3 Weibull Distribution

Definition 3.2: Weibull(λ, α), $\lambda, \alpha > 0$

$$h(x) = \lambda\alpha x^{\alpha-1}, \qquad H(x) = \lambda x^{\alpha}, \qquad S(x) = \exp(-\lambda x^{\alpha}).$$

- Simple yet flexible: hazard increasing ($\alpha > 1$), decreasing ($\alpha < 1$), constant ($\alpha = 1$).
- **Empirical check**: $\ln H(x) = \ln\lambda + \alpha \ln x$, so $\ln \hat{H}(x)$ against $\ln x$ should be linear (slope α).

4 Weibull and the Extreme Value Distribution

Motivation. On the log scale the Weibull is a location–scale model, the basis of AFT regression.

Definition 3.3: Standard extreme value distribution

$$f_W(w) = \exp\{w - e^w\}, \qquad S_W(w) = \exp\{-e^w\}, \qquad -\infty < w < \infty.$$

A special case of the Gompertz family.

Theorem 3.4: Log-linear representation of the Weibull

If $Y \equiv \ln X = \mu + \sigma W$ with W standard extreme value, then $X \sim$ Weibull(λ, α) with

$$\alpha = \frac{1}{\sigma}, \qquad \lambda = e^{-\mu/\sigma} \quad \left(\text{equivalently } \mu = \frac{-\ln\lambda}{\alpha},\ \sigma = \frac{1}{\alpha}\right).$$

Proof [Supplement]. $S_W(w) = \int_w^\infty e^{v-e^v}\,dv = \int_{e^w}^\infty e^{-u}\,du = \exp(-e^w)$ $(u = e^v)$. Then

$$S_X(x) = P(\ln X > \ln x) = P\Big(W > \frac{\ln x - \mu}{\sigma}\Big) = \exp\Big\{-e^{(\ln x-\mu)/\sigma}\Big\} = \exp\Big\{-e^{-\mu/\sigma}x^{1/\sigma}\Big\},$$

i.e. $\exp(-\lambda x^\alpha)$. ■

So a log-linear model with extreme value error gives a Weibull X.

5 Log-normal Distribution

Definition 3.5

If $Y = \ln X \sim N(\mu, \sigma^2)$ then $X \sim$ log-normal(μ, σ^2): $S(x) = 1 - \Phi\Big(\frac{\ln x - \mu}{\sigma}\Big)$.

- Close tie to the normal.
- Hump-shaped hazards: from 0 up to a maximum, then decreasing.
- Criticized because the hazard *always decreases to 0 in the tail.*
- Adequate when large x are not of interest.

6 Log-logistic Distribution

Motivation. The log-logistic allows rising-then-falling hazards in closed form.

Definition 3.6: Log-logistic

If $Y = \ln X = \mu + \sigma W$ with W standard logistic, LG$(0, 1)$:

$$f_W(w) = \frac{e^w}{(1+e^w)^2}, \qquad S_W(w) = \frac{1}{1+e^w},$$

then $Y \sim \text{LG}(\mu, \sigma^2)$ with $f_Y(y) = \dfrac{e^{(y-\mu)/\sigma}}{\sigma\big(1+e^{(y-\mu)/\sigma}\big)^2}$ and $X \sim$ log-logistic.

Derivation: survival function of X

$$S_X(x) = S_W\Big(\frac{\ln x - \mu}{\sigma}\Big) = \frac{1}{1+\exp\left(\frac{\ln x-\mu}{\sigma}\right)} = \frac{1}{1+e^{-\mu/\sigma}x^{1/\sigma}} = \frac{1}{1+\lambda x^\alpha},$$

with $\alpha = 1/\sigma$ and $\lambda = e^{-\mu/\sigma}$.

Theorem 3.7: Log-logistic hazard

$$h_X(x) = \frac{\lambda\alpha x^{\alpha-1}}{1+\lambda x^\alpha}.$$

It is monotone decreasing when $\alpha \le 1$. When $\alpha > 1$ it increases to a maximum at $x^* = \{(\alpha-1)/\lambda\}^{1/\alpha}$ and then decreases to 0 as $x \to \infty$.

Proof [Supplement]. $H(x) = -\ln S(x) = \ln(1+\lambda x^\alpha)$, so $h(x) = H'(x) = \lambda\alpha x^{\alpha-1}/(1+\lambda x^\alpha)$. Differentiating,

$$h'(x) = \lambda\alpha\,\frac{(\alpha-1)x^{\alpha-2}(1+\lambda x^\alpha) - x^{\alpha-1}\lambda\alpha x^{\alpha-1}}{(1+\lambda x^\alpha)^2} = \lambda\alpha\,\frac{x^{\alpha-2}\left[(\alpha-1)-\lambda x^\alpha\right]}{(1+\lambda x^\alpha)^2}.$$

For $\alpha \le 1$ the bracket is negative; for $\alpha > 1$ it changes sign at x^*; and $h(x) \sim \alpha/x \to 0$. ■

Like the log-normal hazard, but in closed form.

7 Gamma Distribution

Definition 3.8: Gamma(λ,β), $\lambda,\beta>0$

$$f(x) = \frac{\lambda^\beta x^{\beta-1}e^{-\lambda x}}{\Gamma(\beta)}, \qquad S(x) = \int_x^\infty \frac{\lambda^\beta s^{\beta-1}e^{-\lambda s}}{\Gamma(\beta)}\,\mathrm{d}s, \qquad h(x) = \frac{x^{\beta-1}e^{-\lambda x}}{\int_x^\infty s^{\beta-1}e^{-\lambda s}\,\mathrm{d}s}.$$

No closed form for S, h in general.

- $\beta = 1$: exponential(λ). $\beta \to \infty$: approximately normal. $\lambda = 1/2$, integer β: $\chi^2_{2\beta}$.
- $\beta > 1$: h increases from $h(0) = 0$ to λ.
- $\beta < 1$: h decreases from $h(0) = \infty$ to λ.
- Less tractable; the Weibull is more popular.

Derivation: why the gamma hazard tends to λ monotonically

Substitute $s = x + u$ in the reciprocal hazard:

$$\frac{1}{h(x)} = \int_x^\infty \left(\frac{s}{x}\right)^{\beta-1} e^{-\lambda(s-x)}\,\mathrm{d}s = \int_0^\infty \left(1+\frac{u}{x}\right)^{\beta-1} e^{-\lambda u}\,\mathrm{d}u.$$

As $x \to \infty$, $1/h(x) \to \int_0^\infty e^{-\lambda u}\,\mathrm{d}u = 1/\lambda$. For $\beta > 1$, $(1+u/x)^{\beta-1}$ decreases in x, so h increases; for $\beta < 1$ the reverse.

8 Regression Models for Survival Data

Heterogeneity requires covariate adjustment, e.g. for groups imbalanced in age, tumour size, positive nodes or hormonal status in non-randomized studies.

8.1 Approach 1: Log-linear / accelerated failure time (AFT) model

Definition 3.9: Log-linear model

$$Y \equiv \ln X = \mu + \boldsymbol{\gamma}'\mathbf{z} + \sigma W,$$

with $W \sim F$ known and $\mu, \sigma, \boldsymbol{\gamma}$ unknown.

Theorem 3.10: Log-linear ⇒ AFT

Let $X_0 = \exp(\mu + \sigma W)$ with survival S_0. Then

$$S(x \mid \mathbf{z}) = S_0(x\, e^{-\boldsymbol{\gamma}'\mathbf{z}}), \qquad h(x \mid \mathbf{z}) = \eta\, h_0(\eta x), \quad \eta = e^{-\boldsymbol{\gamma}'\mathbf{z}}.$$

Proof [Supplement].

$$\begin{aligned} P(X > x \mid \mathbf{z}) &= P(Y > \ln x \mid \mathbf{z}) = P(\mu + \sigma W > \ln x - \boldsymbol{\gamma}'\mathbf{z}) \\ &= P\big(\exp(\mu + \sigma W) > x \exp(-\boldsymbol{\gamma}'\mathbf{z})\big) \\ &= S_0(xe^{-\boldsymbol{\gamma}'\mathbf{z}}). \end{aligned}$$

$S(x \mid \mathbf{z}) = S_0(\eta x)$ gives $f(x \mid \mathbf{z}) = \eta f_0(\eta x)$, so

$$h(x \mid \mathbf{z}) = \eta f_0(\eta x)/S_0(\eta x) = \eta h_0(\eta x).$$

■

- Covariates rescale time: **accelerated** if $\eta = e^{-\boldsymbol{\gamma}'\mathbf{z}} > 1$, **decelerated** if $\eta < 1$.
- Signs are **opposite**: positive γ lengthens survival ($\eta < 1$).

Derivation: interpretation of γ

Since $X = e^{\mu + \boldsymbol{\gamma}'\mathbf{z}} e^{\sigma W}$,

$$\frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \frac{e^{\mu + \boldsymbol{\gamma}'\mathbf{z}_1}\mathrm{E}(e^{\sigma W})}{e^{\mu + \boldsymbol{\gamma}'\mathbf{z}_2}\mathrm{E}(e^{\sigma W})} = e^{\boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2)} \implies \ln \frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2).$$

γ is the log ratio of mean (or any quantile of) survival times per unit increase in z.

Definition 3.11: Three choices of F

1. $W \sim N(0,1)$: $\ln X \sim N(\mu + \boldsymbol{\gamma}'\mathbf{z}, \sigma^2)$, i.e. X is **log-normal**.
2. W standard extreme value, $f_W(w) = \exp(w - e^w)$: $X \sim \text{Weibull}(\lambda, \alpha)$ with $\alpha = 1/\sigma$ and $\lambda = \exp\{-(\frac{\mu}{\sigma} + \frac{\boldsymbol{\gamma}'\mathbf{z}}{\sigma})\}$.
3. W standard logistic: the **log-logistic** regression model.

8.2 Approach 2: Multiplicative (proportional hazards) model

Definition 3.12: Multiplicative hazards model

$$h(x \mid \mathbf{z}) = h_0(x)\, C(\boldsymbol{\beta}'\mathbf{z}),$$

with baseline $h_0(x) \geq 0$ parametric or unspecified; $C(\boldsymbol{\beta}'\mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}}$ gives the **Cox proportional hazards model**.

- β is the **log hazard ratio**, constant over time: $h(x \mid \mathbf{z}_1)/h(x \mid \mathbf{z}_2) = e^{\boldsymbol{\beta}'(\mathbf{z}_1 - \mathbf{z}_2)}$.
- Integrating, $H(x \mid \mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)$ and $S(x \mid \mathbf{z}) = \exp\{-e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)\} = \{S_0(x)\}^{e^{\boldsymbol{\beta}'\mathbf{z}}}$.
- **Graphical check**: $\ln[-\ln S(x \mid \mathbf{z})] = \boldsymbol{\beta}'\mathbf{z} + \ln[-\ln S_0(x)]$, so log-cumulative-hazard curves should be *parallel*.

Example 3.13: Weibull baseline under PH

If $h_0(x) = \alpha\lambda x^{\alpha-1}$ then $h(x \mid \mathbf{z}) = \alpha\lambda x^{\alpha-1}\exp(\boldsymbol{\beta}'\mathbf{z})$ and

$$S(x \mid \mathbf{z}) = \exp\big[-\lambda\exp(\boldsymbol{\beta}'\mathbf{z})x^{\alpha}\big], \quad \text{i.e.} \quad X \mid \mathbf{z} \sim \text{Weibull}(\lambda e^{\boldsymbol{\beta}'\mathbf{z}}, \alpha).$$

Theorem 3.14: Weibull is closed under both AFT and PH

The Weibull is the only continuous distribution closed under both AFT and PH.

Derivation: matching the two parametrizations

Under AFT with Weibull baseline, $h(x \mid \mathbf{z}) = \eta\,\alpha\lambda(\eta x)^{\alpha-1} = \eta^{\alpha}\cdot\alpha\lambda x^{\alpha-1}$: PH with hazard ratio $\eta^{\alpha} = e^{-\alpha\boldsymbol{\gamma}'\mathbf{z}}$, so

$$\boldsymbol{\beta} = -\alpha\boldsymbol{\gamma} = -\boldsymbol{\gamma}/\sigma.$$

8.3 Approach 3: Additive hazards model

Definition 3.15: Additive hazards model

$$h(x \mid \mathbf{z}) = h_0(x) + \sum_{j=1}^{p} z_j(x)\beta_j(x).$$

- Covariate effects may vary over time.
- $\beta_j(x)$ must be constrained so that $h(x \mid \mathbf{z}) \geq 0$.

Lecture 4. Formal Introduction to Censoring and Truncation

1 Right Censoring

Definition 4.1: Right-censored data

X is only known to exceed some value. With event time X and censoring time C we observe

$$(T, \delta), \qquad T = \min(X, C), \qquad \delta = I(X < C) \text{ (or } I(X \leq C)),$$

for $i = 1, \ldots, n$.

Some types of right censoring give information on C even when $X < C$.

1.1 Type I censoring

The event is observed only before a *pre-specified* time.

- **Fixed Type I**: $C_i = \tau$ for all i, e.g. mice followed for a fixed period after carcinogen exposure, survivors censored at study end.
- **Progressive Type I**: C fixed in advance with finitely many distinct values, e.g. time to a given weight loss with mice in cages 1–2 sacrificed after week 1 and cages 3–4 after week 2.
- **Generalized Type I**: C_i individual but known in advance, e.g. *staggered entry* in a Phase I study: C_i = (end of study) − (entry time), assuming no dropout.

1.2 Type II censoring

- **Simple Type II**: continue until the first r of n failures; r and $n - r$ are *fixed*, $C = X_{(r)}$ is *random*.
- **Progressive Type II**: after r_1 failures remove $n_1 - r_1$ of the $n - r_1$ survivors; after the next r_2, remove $n_2 - r_2$; and so on.

Remark 4.2: Type I vs. Type II

Type I: C predetermined. Type II: C is the random time of a predetermined number of events. This matters for the likelihood.

1.3 Random censoring

Human studies: $C \sim G(t)$, not fixed, possibly unknown.

- Administrative censoring at study end (like Type I).
- Drop-out before study end (random).
- In practice, a mixture.

2 Left, Double and Interval Censoring

Definition 4.3: Left censoring

X is only known to be less than C_l; exact iff $X \geq C_l$. Data: $T = \max(X, C_l)$, $\delta = I(X \geq C_l)$.

Definition 4.4: Double censoring

With left censoring time C_l and right censoring time C_r ($C_l < C_r$):

$$T = \max\{\min(X, C_r),\, C_l\}, \qquad \delta = \begin{cases} 1, & C_l \leq X \leq C_r \text{ (exact)} \\ -1, & X < C_l \text{ (left censored)} \\ 0, & X > C_r \text{ (right censored)} \end{cases}$$

Remark 4.5: Double vs. interval censoring

Interval: each observation lies in an interval $L < X \leq R$, none exact. **Double**: each is exact, right or left censored.

3 Truncation

Definition 4.6: Truncation

Only event times in a window (Y_L, Y_R) are observed:

- $Y_R = \infty$: **left truncation** ($Y_L < X$);
- $Y_L = 0$: **right truncation** ($X \leq Y_R$).

Inference is *conditional*.

Example 4.7: Breast cancer onset (classification exercise)

8 healthy 50-year-old women had yearly mammograms for 10 years. Onset ages:

$$(55, 56],\ (58, 59],\ (52, 53],\ (59, 60],\ > 60,\ > 60,\ > 60,\ > 60.$$

Answer. Four detected cases are **interval censored**, four undetected ones **right censored** at 60, and all are **left truncated** at 50 (only cancer-free women enrolled).

4 Likelihood Construction

4.1 Contributions of each observation type

X continuous with f, F, S; C_r, C_l censoring points, $(L, R]$ censoring interval, Y truncation point.

Observation	Likelihood contribution
Exact lifetime	$f(x)$
Right censored	$S(C_r) = P(X > C_r)$
Left censored	$1 - S(C_l) = P(X < C_l)$
Interval censored	$S(L) - S(R) = P(L < X \le R)$
Left truncated	$f(x)/S(Y_L)$
Right truncated	$f(x)/[1 - S(Y_R)]$
Interval truncated	$f(x)/[S(Y_L) - S(Y_R)]$

Truncated contributions are conditional densities, e.g. $f(x \mid X > Y_L) = f(x)/S(Y_L)$.

Theorem 4.8: General censored-data likelihood

$$L \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_{r,i}) \prod_{i \in LC} F(C_{l,i}) \prod_{i \in IC} \{S(L_i) - S(R_i)\},$$

with D exact times and $RC/LC/IC$ the censored observations.

Truncation. Left truncated: $f(x_i) \to f(x_i)/S(Y_i)$, $S(C_{r,i}) \to S(C_{r,i})/S(Y_i)$. Right truncated: $f(x_i) \to f(x_i)/[1 - S(Y_i)]$.

4.2 Type I right censoring

Theorem 4.9

Observing (T_i, δ_i) with $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and fixed C_i,

$$L = \prod_{i=1}^{n} \Pr(t_i, \delta_i) = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i} = \prod_{i=1}^{n} h(t_i)^{\delta_i} e^{-H(t_i)}.$$

Proof [Supplement]. With C_i fixed, $\delta_i = 0$ means $X_i > C_i = t_i$, probability $S(t_i)$; $\delta_i = 1$ gives density $f(t_i)$. With $f = hS$, $S = e^{-H}$: $f^{\delta} S^{1-\delta} = (hS)^{\delta} S^{1-\delta} = h^{\delta} S = h^{\delta} e^{-H}$. ■

Example 4.10: Exponential with right censoring

$X \sim \text{Exp}(\lambda)$, $T = \min(X, C)$, $\delta = I(X < C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

With $h(t) = \lambda$ and $H(t) = \lambda t$,

$$L(\lambda) = \prod_{i=1}^{5} \lambda^{\delta_i} e^{-\lambda t_i} = \lambda^{\sum \delta_i} e^{-\lambda \sum t_i} = \lambda^3 e^{-3.25\lambda}.$$

$\ell'(\lambda) = 3/\lambda - 3.25 = 0$ gives $\hat{\lambda} \approx 0.923$; generally $\hat{\lambda} = \sum \delta_i / \sum t_i$ (events / time at risk).

4.3 Type II censoring

Theorem 4.11

With n i.i.d. $X_i \sim f$, observing the r smallest order statistics $X_{(1)} < \cdots < X_{(r)}$ and $n-r$ values censored at $X_{(r)}$,

$$L = \frac{n!}{(n-r)!} \prod_{i=1}^{r} f(x_{(i)}) \left\{S(x_{(r)})\right\}^{n-r}.$$

Derivation: where the constant comes from

$n!/(n-r)!$ ordered choices of the r failing units, each contributing $f(x_{(i)})$, and $n-r$ survivors beyond $x_{(r)}$. The constant is parameter-free, so the kernel matches Theorem 4.8.

Progressive Type II (two stages: r_1 failures, $n_1 - r_1$ removed, r_2 failures $X^*_{(i)}$, the rest removed):

$$L \propto \prod_{i=1}^{r_1} f(x_{(i)}) \left\{S(x_{(r_1)})\right\}^{n_1-r_1} \prod_{i=1}^{r_2} f(x^*_{(i)}) \left\{S(x^*_{(r_2)})\right\}^{n-n_1-r_2} \propto \prod_{i\in D} f(x_i) \prod_{i\in RC} S(C_i).$$

4.4 Random censoring

$X_i \sim f, S$ and $C_i \sim g, G$ random and **independent**; $T_i = \min(X_i, C_i)$, $\delta_i = I(X_i \le C_i)$.

Theorem 4.12

$$L = \prod_{i=1}^{n} \left\{f(t_i)G(t_i)\right\}^{\delta_i} \left\{g(t_i)S(t_i)\right\}^{1-\delta_i} = \underbrace{\prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}}_{L_X} \underbrace{\prod_{i=1}^{n} g(t_i)^{1-\delta_i} G(t_i)^{\delta_i}}_{L_C}.$$

Proof [Supplement]. $P(T \in \mathrm{d}t, \delta = 1) = P(X \in \mathrm{d}t,\ C \ge t) = f(t)G(t)\,\mathrm{d}t$ and $P(T \in \mathrm{d}t, \delta = 0) = g(t)S(t)\,\mathrm{d}t$ by independence. ■

If the law of C is free of θ (**non-informative censoring**), L_C is constant and

$$L \propto L_X = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i},$$

as under Type I censoring.

5 Remarks

- Used mainly for parametric models (K&M Ch. 12) and behind the Cox *partial likelihood* (Ch. 8–9).
- Rarely written explicitly, but Theorem 4.8 explains each data component's contribution.

6 Selected exercises

Motivation. K&M Section 3.7: a likelihood with truncation and two kinds of censoring, and the exact law of the exponential MLE under random censoring.

Exercise 4.1 (K&M 3.2: periodic exams with delayed entry)

Disease-free women enrolled from 1970 were followed 30 years for age X at breast cancer onset, examined every 3 years (onset detected only at exams). Classify, and give the likelihood contribution:
(a) enrolled at age 30, never developed breast cancer during the study;
(b) enrolled at age 40, diagnosed at the fifth exam after enrollment;
(c) enrolled at age 50, died of an unrelated cause at age 61 without a diagnosis;
(d) enrolled at age 42, moved away at age 55 without a diagnosis.
(e) Write the likelihood for these four women.

Solution. Each woman is *left truncated* at entry age v (contribution divided by $S(v)$); onset is known only up to exams at $v+3, v+6, \ldots$

(a) Exams $33, \ldots, 60$ negative: right censored at 60; $S(60)/S(30)$.

(b) Exam 4 (age 52) negative, exam 5 (55) positive: interval censored; $\{S(52) - S(55)\}/S(40)$.

(c) Exams $53, 56, 59$ negative; death at 61 ends follow-up, onset could be unseen after 59: right censored at 59; $S(59)/S(50)$. (If onset before death would surely be noticed, $S(61)/S(50)$.)

(d) Exams $45, \ldots, 54$ negative, then lost: $S(54)/S(42)$.

(e) With independent censoring and truncation,

$$L = \frac{S(60)}{S(30)} \cdot \frac{S(52) - S(55)}{S(40)} \cdot \frac{S(59)}{S(50)} \cdot \frac{S(54)}{S(42)}.$$

■

Exercise 4.2 (K&M 3.8: exponential lifetimes, exponential censoring)

Let $X \sim \text{Exp}(\lambda)$ and $C \sim \text{Exp}(\theta)$ be independent, $T = \min(X, C)$, $\delta = I(X \le C)$.
(a) Find $P(\delta = 1)$.
(b) Find the distribution of T.
(c) Show that δ and T are independent.
(d) For a random sample (T_i, δ_i), $i = 1, \ldots, n$, show that the MLE of λ is $\hat{\lambda} = \sum_i \delta_i / \sum_i T_i$, and use (a)–(c) to find its mean ($n \ge 2$) and variance ($n \ge 3$).

Solution.

(a)–(c) For $t \ge 0$,

$$P(\delta = 1, T > t) = P(t < X \le C) = \int_t^\infty \lambda e^{-\lambda s} e^{-\theta s}\, \mathrm{d}s = \frac{\lambda}{\lambda + \theta} e^{-(\lambda+\theta)t}.$$

$t = 0$ gives $P(\delta = 1) = \lambda/(\lambda + \theta)$; $P(T > t) = e^{-(\lambda+\theta)t}$, so $T \sim \text{Exp}(\lambda + \theta)$; and $P(\delta = 1, T > t) = P(\delta = 1)P(T > t)$ (hence also for $\delta = 0$): $\delta \perp\!\!\!\perp T$.

(d)

Step 1 (MLE). $L = \lambda^D e^{-\lambda G} \cdot \theta^{n-D} e^{-\theta G}$, $D = \sum \delta_i$, $G = \sum T_i$; $\partial_\lambda \log L = D/\lambda - G$ changes sign at D/G (if $D = 0$, $\hat{\lambda} = 0$).

Step 2 (distributions). By (c) all δ_i, T_i are independent: $D \sim \text{Bin}(n, p)$, $p = \lambda/r$, $r = \lambda + \theta$; $G \sim \text{Gamma}(r, n)$ (shape n, rate r); $D \perp\!\!\!\perp G$.

Step 3 (inverse moments of G). For $k < n$,

$$\mathrm{E}(G^{-k}) = \int_0^\infty g^{-k} \frac{r^n g^{n-1} e^{-rg}}{\Gamma(n)}\, \mathrm{d}g = \frac{r^k\, \Gamma(n-k)}{\Gamma(n)},$$

so $\mathrm{E}(G^{-1}) = r/(n-1)$ and $\mathrm{E}(G^{-2}) = r^2/\{(n-1)(n-2)\}$.

Step 4 (mean). $\mathrm{E}\hat{\lambda} = np \cdot r/(n-1) = n\lambda/(n-1)$, slightly biased upward.

Step 5 (variance). $\mathrm{E}(D^2) = np(1-p) + n^2p^2$, $r^2p(1-p) = \lambda\theta$, $r^2p^2 = \lambda^2$:

$$\mathrm{E}\hat{\lambda}^2 = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)},$$

$$\mathrm{Var}\hat{\lambda} = \frac{n\lambda\theta + n^2\lambda^2}{(n-1)(n-2)} - \frac{n^2\lambda^2}{(n-1)^2} = \frac{n\lambda\theta}{(n-1)(n-2)} + \frac{n^2\lambda^2}{(n-1)^2(n-2)}.$$

As $n \to \infty$, $n\mathrm{Var}\hat{\lambda} \to \lambda^2/p$, the inverse Fisher information per observation. ■

Lecture 5. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 1)

1 Setting

X is continuous with survival S and cumulative hazard H, under **non-informative right censoring**:

$$\{(T_i, \delta_i),\ i = 1, \dots, n\}, \qquad T_i = \min(X_i, C_i), \quad \delta_i = I(X_i \le C_i).$$

Goal: estimate S and H nonparametrically, in four steps (discrete/continuous $\times$ without/with censoring).

2 Building Up the Estimator

2.1 Discrete X without censoring

Let X have support $0 < \tilde{x}_1 < \cdots < \tilde{x}_M < \infty$, $\tilde{x}_0 = 0$. By the product-limit form (Lecture 2),

$$\begin{aligned} S(x) = P(X > x) &= \prod_{\tilde{x}_j \le x} \frac{S(\tilde{x}_j)}{S(\tilde{x}_{j-1})} = \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X > \tilde{x}_{j-1}) \\ &= \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X \ge \tilde{x}_j) = \prod_{\tilde{x}_j \le x} [1 - h(\tilde{x}_j)], \qquad h(\tilde{x}_j) = P(X = \tilde{x}_j \mid X \ge \tilde{x}_j). \end{aligned}$$

(Third equality: no mass between $\tilde{x}_{j-1}$ and $\tilde{x}_j$.) For i.i.d. $X_1, \dots, X_n$, estimate

$$\hat{h}(\tilde{x}_j) = \frac{\#\{X_i = \tilde{x}_j\}}{\#\{X_i \ge \tilde{x}_j\}} = \frac{\#\text{ deaths at } \tilde{x}_j}{\#\text{ deaths at } \tilde{x}_j, \tilde{x}_{j+1}, \dots, \tilde{x}_M}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} [1 - \hat{h}(\tilde{x}_j)].$$

These are the MLEs.

2.2 Discrete X with right censoring

With $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and $X_i \perp\!\!\!\perp C_i$:

$$\begin{aligned} P(T_i = \tilde{x}_j, \delta_i = 1 \mid T_i \ge \tilde{x}_j) &= \frac{P(T_i = \tilde{x}_j, \delta_i = 1)}{P(T_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j,\ C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j,\ C_i \ge \tilde{x}_j)} \\ &= \frac{P(X_i = \tilde{x}_j)P(C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)P(C_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)} = h(\tilde{x}_j). \end{aligned}$$

So the hazard is identifiable from the *observed* data:

$$\hat{h}(\tilde{x}_j) = \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} \Big[1 - \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}\Big].$$

Derivation: $\hat{h}$ is the MLE under censoring

Under independent censoring the likelihood factorizes over support points: of the Y_j at risk at $\tilde{x}_j$, d_j die (probability h_j each), the rest survive past it; censored subjects contribute survival

to their censoring time. Thus

$$L(h_1,\dots,h_M) \propto \prod_j h_j^{d_j}(1-h_j)^{Y_j-d_j}, \qquad \frac{\partial \log L}{\partial h_j} = \frac{d_j}{h_j} - \frac{Y_j-d_j}{1-h_j} = 0 \;\Rightarrow\; \hat{h}_j = \frac{d_j}{Y_j}.$$

By invariance $\hat{S}(x) = \prod_{\tilde{x}_j \le x}(1-\hat{h}_j)$ is the MLE of S.

2.3 Continuous X

Without censoring, use the empirical survival function

$$\hat{S}(t) = 1 - \mathbb{F}_n(t) = 1 - \frac{1}{n}\sum_{i=1}^{n} I(X_i \le t).$$

With censoring: the estimator should not jump at censoring times (no event there), and a censored observation counts in Y_j up to its censoring time, then leaves.

3 The Kaplan–Meier (Product-Limit) Estimator

Motivation. Kaplan–Meier multiplies conditional survival probabilities at the failure times, each estimated from the risk set.

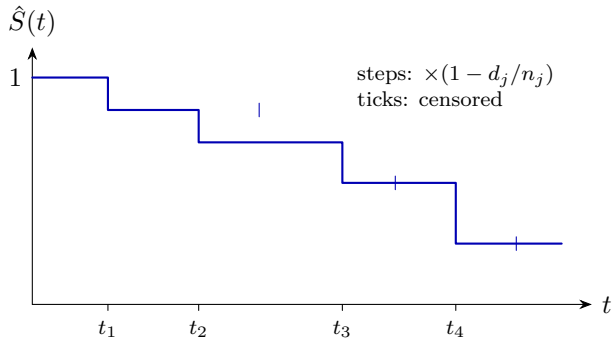


Figure. *KM drops only at failure times t_j, by $1-d_j/n_j$; censorings (ticks) only shrink the risk set.*

Definition 5.1: Notation

- $t_1 < t_2 < \cdots < t_D$: the D distinct death times;
- $d_j = \sum_{i=1}^n I(T_i = t_j, \delta_i = 1)$: number of deaths at t_j;
- $Y_j = \sum_{i=1}^n I(T_i \ge t_j)$: number at risk just before t_j.

Theorem 5.2: Kaplan–Meier estimator

$$\hat{S}(t) = \begin{cases} 1, & t < t_1, \\ \displaystyle\prod_{t_j \le t}\Big(1 - \frac{d_j}{Y_j}\Big), & t_1 \le t. \end{cases}$$

Remark 5.3: Ties between deaths and censorings

A censoring tied with a death is taken to occur just *after* it, so it counts in Y_j.

Theorem 5.4: Greenwood's variance estimator

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}.$$

Derivation: delta-method derivation (Cox & Oakes, 1984)

With $\hat{h}_j = d_j/Y_j$, $d_j \mid Y_j \sim \mathrm{Bin}(Y_j, h_j)$ and the $\hat{h}_j$ approximately uncorrelated,

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j}.$$

With $\ln \hat{S}(t) = \sum_{t_j \le t} \ln(1-\hat{h}_j)$ and the delta method ($g(u) = \ln(1-u)$, $g'(u) = -1/(1-u)$):

$$\begin{aligned}
\mathrm{Var}\{\ln \hat{S}(t)\} &\approx \sum_{t_j \le t} \mathrm{Var}\{\ln(1-\hat{h}_j)\} \approx \sum_{t_j \le t} \Big(\frac{1}{1-\hat{h}_j}\Big)^2 \mathrm{Var}(\hat{h}_j) \\
&\approx \sum_{t_j \le t} \Big(\frac{1}{1-\hat{h}_j}\Big)^2 \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \le t} \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \le t} \frac{d_j/Y_j}{Y_j(Y_j-d_j)/Y_j} \\
&= \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j-d_j)}.
\end{aligned}$$

A second delta step: $\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \,\mathrm{Var}\{\ln \hat{S}(t)\}$.

4 The Nelson–Aalen Estimator

Motivation. Summing the failure rates d_j/Y_j estimates the cumulative hazard, with better small-sample behaviour.

Theorem 5.5: Nelson–Aalen estimator and Aalen's variance

$$\tilde{H}(t) = \begin{cases} 0, & t < t_1, \\ \displaystyle\sum_{t_j \le t} \frac{d_j}{Y_j}, & t_1 \le t, \end{cases} \qquad \sigma^2_H(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Remark 5.6: Intuition

Each death time adds the hazard increment d_j/Y_j. Treating d_j as Poisson with mean $Y_j h_j$, $\mathrm{Var}(d_j/Y_j) \approx d_j/Y_j^2$.

4.1 Alternative estimators

- Alternative for H: $\hat{H}(t) = -\log \hat{S}(t)$.

- Alternative for S: $\tilde{S}(t) = \exp\{-\tilde{H}(t)\}$ (Breslow / Fleming–Harrington).
- $1 - x \le e^{-x}$ gives $\hat{S}(t) \le \tilde{S}(t)$, i.e. $\tilde{H}(t) \le \hat{H}(t)$.
- "Product-limit": as $n \to \infty$ the number of factors grows and the product tends to a *product integral* over the support.

5 Worked Example (Small Sample)

Example 5.7: $n = 5$

Right-censored continuous X:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

Ordered: $0.25^+,\ 0.5,\ 0.75,\ 0.75^+,\ 1$; at 0.75 the censored subject is at risk.

t_j	d_j	Y_j	$1 - d_j/Y_j$	$\hat{S}(t_j)$	d_j/Y_j	$\tilde{H}(t_j)$	$\tilde{S}(t_j) = e^{-\tilde{H}}$
0.5	1	4	3/4	0.750	0.250	0.250	0.779
0.75	1	3	2/3	0.500	0.333	0.583	0.558
1	1	1	0	0.000	1.000	1.583	0.205

Greenwood: $\hat{V}[\hat{S}(0.5)] = 0.75^2 \cdot \frac{1}{12} = 0.0469$, $\hat{V}[\hat{S}(0.75)] = 0.5^2(\frac{1}{12} + \frac{1}{6}) = 0.0625$; undefined at $t = 1$ ($Y_j = d_j$).

6 Remarks

- $\hat{S}$, $\tilde{H}$, $\hat{H}$ are right-continuous step functions.
- $\hat{S}$, $\tilde{S}$ are consistent for $t \le \tau$ with $P(T \ge \tau) > 0$, and asymptotically equivalent.
- Both are defined only up to $t_{\max} = \max T_i$; if the largest observation is censored, $\hat{S}$ does not reach 0. Extensions beyond $t_{\max}$:
  - Efron (1967): $\hat{S}(t) = 0$;
  - Gill (1980): $\hat{S}(t) = \hat{S}(t_{\max})$;
  - Brown, Hollander & Kowar (1974): $\hat{S}(t) = \exp\{t \log[\hat{S}(t_{\max})]/t_{\max}\}$.

7 The 6-MP Leukaemia Trial

Data. 42 children with acute leukaemia in remission, matched in pairs by remission status, randomized within pairs to 6-mercaptopurine (6-MP) or placebo, and followed to relapse or study end. Weeks to relapse, 6-MP ($^+$ = censored):

$$6,\ 6,\ 6,\ 6^+,\ 7,\ 9^+,\ 10,\ 10^+,\ 11^+,\ 13,\ 16,\ 17^+,\ 19^+,\ 20^+,\ 22,\ 23,\ 25^+,\ 32^+,\ 32^+,\ 34^+,\ 35^+.$$

Placebo (all observed): $1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23$.

Example 5.8: Kaplan–Meier estimate for the 6-MP group

t_j	d_j	Y_j	$\hat{S}(t)=\prod(1-d_j/Y_j)$	$\sum \frac{d_j}{Y_j(Y_j-d_j)}$	$\hat{S}(t)^2\sum\cdots$
6	3	21	$[1-\frac{3}{21}]=0.857$	$\frac{3}{21\times18}=0.0079$	0.0058
7	1	17	$0.857(1-\frac{1}{17})=0.807$	$0.0079+\frac{1}{17\times16}=0.0116$	0.0076
10	1	15	$0.807(1-\frac{1}{15})=0.753$	$0.0116+\frac{1}{15\times14}=0.0164$	0.0093
13	1	12	$0.753(1-\frac{1}{12})=0.690$	$0.0164+\frac{1}{12\times11}=0.0240$	0.0114
16	1	11	$0.690(1-\frac{1}{11})=0.628$	$0.0240+\frac{1}{11\times10}=0.0330$	0.0130
22	1	7	$0.628(1-\frac{1}{7})=0.538$	$0.0330+\frac{1}{7\times6}=0.0569$	0.0164
23	1	6	$0.538(1-\frac{1}{6})=0.448$	$0.0569+\frac{1}{6\times5}=0.0902$	0.0181

Censorings shrink Y_j without a jump, e.g. 9^+ between $t=7$ ($Y=17$) and $t=10$ ($Y=15$).

Example 5.9: Nelson–Aalen estimate for the 6-MP group

t	$\tilde{H}(t)=\sum d_j/Y_j$	$\sigma_H^2=\sum d_j/Y_j^2$	SE
$0\le t<6$	0	0	0
$6\le t<7$	$3/21=0.1428$	$3/21^2=0.0068$	0.0825
$7\le t<10$	$0.1428+1/17=0.2017$	$0.0068+1/17^2=0.0103$	0.1015
$10\le t<13$	$0.2017+1/15=0.2683$	$0.0103+1/15^2=0.0147$	0.1212
$13\le t<16$	$0.2683+1/12=0.3517$	$0.0147+1/12^2=0.0217$	0.1473
$16\le t<22$	$0.3517+1/11=0.4426$	$0.0217+1/11^2=0.0299$	0.1729
$22\le t<23$	$0.4426+1/7=0.5854$	$0.0299+1/7^2=0.0503$	0.2243
$23\le t<35$	$0.5854+1/6=0.7521$	$0.0503+1/6^2=0.0781$	0.2795

Comparison. $\tilde{S}=e^{-\tilde{H}}$ lies slightly *above* KM (K&M Fig. 4.1A), most in the tail (at $t=23$: $\hat{S}=0.448$, $\tilde{S}=0.471$).

8 Selected exercises

Motivation. Without censoring KM is the empirical survival function; KM redistributes censored mass to the right; Nelson–Aalen matches the moments of H at order statistics.

Exercise 5.1 (No censoring: KM and Greenwood reduce to binomial quantities)

Suppose no observation is censored ($\delta_i=1$ for all i; ties allowed).
(a) Show that $\hat{S}(t)=1-\mathbb{F}_n(t)$ for all t.
(b) Show that Greenwood's formula reduces to $\hat{V}[\hat{S}(t)]=\hat{S}(t)\{1-\hat{S}(t)\}/n$ whenever $\hat{S}(t)>0$.

Solution [Supplement].

(a)

Step 1. With $Y_{D+1}=0$: without censoring $Y_{j+1}=Y_j-d_j$, $Y_1=n$.

Step 2. So $1-d_j/Y_j=Y_{j+1}/Y_j$, and for $t_k\le t<t_{k+1}$ ($t_{D+1}=\infty$)

$$\hat{S}(t)=\prod_{j=1}^{k}\frac{Y_{j+1}}{Y_j}=\frac{Y_{k+1}}{n}=\frac{\#\{i:T_i>t\}}{n}=1-\mathbb{F}_n(t).$$

(For $t < t_1$ both are 1.)

(b) If $\hat S(t) > 0$, $Y_{k+1} > 0$, and

$$\frac{d_j}{Y_j(Y_j - d_j)} = \frac{Y_j - Y_{j+1}}{Y_j Y_{j+1}} = \frac{1}{Y_{j+1}} - \frac{1}{Y_j}, \qquad \sum_{j=1}^{k} \frac{d_j}{Y_j(Y_j - d_j)} = \frac{1}{Y_{k+1}} - \frac{1}{n}.$$

With $\hat S = Y_{k+1}/n$,

$$\hat V[\hat S(t)] = \frac{Y_{k+1}^2}{n^2}\Big(\frac{1}{Y_{k+1}} - \frac{1}{n}\Big) = \frac{1}{n} \cdot \frac{Y_{k+1}}{n}\Big(1 - \frac{Y_{k+1}}{n}\Big) = \frac{\hat S(t)\{1 - \hat S(t)\}}{n},$$

the binomial variance estimate. ■

Exercise 5.2 (Self-consistency of the Kaplan–Meier estimator)

Let t be such that $\hat S(t) > 0$. Show that

$$\hat S(t) = \frac{1}{n}\Big[\#\{i : T_i > t\} + \sum_{i:\,\delta_i = 0,\ T_i \le t} \frac{\hat S(t)}{\hat S(T_i)}\Big].$$

I.e. each subject censored at $T_i \le t$ counts as alive at t with probability $\hat P(X > t \mid X > T_i)$.

Solution [Supplement].

Step 1. For $s \le t$ let $R(s) = \#\{i : T_i > s\}$, $A(s) = \sum_{i:\delta_i=0, T_i \le s} 1/\hat S(T_i)$ (well defined, $\hat S(s) \ge \hat S(t) > 0$), and

$$\varphi(s) = \frac{R(s)}{\hat S(s)} + A(s).$$

The claim is $\varphi(t) = n$.

Step 2. R, $\hat S$, A change only at observed times; for $s < \min_i T_i$, $\varphi(s) = n$.

Step 3. At an observed $s \le t$ with $Y = R(s-)$ at risk, d deaths and c censorings: $R(s) = Y - d - c$, $A(s) = A(s-) + c/\hat S(s)$, $\hat S(s) = \hat S(s-)(1 - d/Y)$. Hence

$$\varphi(s) - \varphi(s-) = \frac{Y - d - c}{\hat S(s)} + \frac{c}{\hat S(s)} - \frac{Y}{\hat S(s-)} = \frac{(Y - d) - Y(1 - d/Y)}{\hat S(s)} = 0.$$

(The tie convention puts the c censored subjects in Y.)

Step 4. So $\varphi \equiv n$ on $(-\infty, t]$.

Efron (1967): this "redistribute-to-the-right" property characterizes $\hat S$; it underlies Turnbull's algorithm (Lecture 8). ■

Exercise 5.3 (Nelson–Aalen at the order statistics)

Let $X_1, \dots, X_n$ be i.i.d. with continuous F, no censoring, and order statistics $X_{(1)} < \cdots < X_{(n)}$.

(a) Show that $\tilde H(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{n-i+1}$ and $\sigma_H^2(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{(n-i+1)^2}$.

(b) Show that $\mathrm{E}\{H(X_{(k)})\} = \tilde H(X_{(k)})$ and $\mathrm{Var}\{H(X_{(k)})\} = \sigma_H^2(X_{(k)})$.

Solution [Supplement].

(a) No ties a.s., so $d_i = 1$, $Y_i = n - i + 1$ at $X_{(i)}$; use Theorem 5.5.

(b)

Step 1. $S(X) \sim U(0,1)$, so $P\{H(X) > u\} = P\{S(X) < e^{-u}\} = e^{-u}$: $E_i = H(X_i)$ are i.i.d. Exp(1).

Step 2. H is nondecreasing: $H(X_{(k)}) = E_{(k)}$.

Step 3. $(E_{(1)}, \ldots, E_{(n)})$ has density $n! \exp(-\sum_i e_i)$ on $0 < e_1 < \cdots < e_n$. The spacings $D_i = E_{(i)} - E_{(i-1)}$ $(E_{(0)} = 0)$ have unit Jacobian and $\sum_i e_i = \sum_i (n-i+1)d_i$, so their density is

$$\prod_{i=1}^{n} (n-i+1)\, e^{-(n-i+1)d_i}, \qquad d_i > 0,$$

i.e. independent $D_i \sim \text{Exp}(n-i+1)$.

Step 4. $E_{(k)} = \sum_{i \le k} D_i$ has mean $\sum_{i \le k} (n-i+1)^{-1}$ and variance $\sum_{i \le k} (n-i+1)^{-2}$; compare (a). ■

Lecture 6. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 2)

1 Recap

$$\hat{S}(t) = \prod_{t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}; \qquad \tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j}, \quad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Define

$$\sigma_S^2(t) = \frac{\hat{V}[\hat{S}(t)]}{\hat{S}^2(t)} = \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

so $\hat{S}(t)\sigma_S(t)$ is the Greenwood SE.

2 Pointwise Confidence Intervals for $S(t)$

Motivation. Greenwood gives a direct interval that may leave $[0, 1]$; log and log-log transforms fix this.

2.1 Linear interval

Definition 6.1: Linear CI

$$\Big[\hat{S}(t) - Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t), \quad \hat{S}(t) + Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t)\Big].$$

Symmetric; can fall outside $[0, 1]$.

2.2 Transformed intervals: the general recipe

Derivation: delta method for a transformation g

Let g be known, monotone and differentiable. By the delta method, For known monotone differentiable g, $\mathrm{SE}\{g(\hat{S})\} \approx |g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)$, so a $100(1-\alpha)\%$ CI for $g(S(t))$ is

$$g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t),$$

and for $S(t)$:

$$g^{-1}\Big\{g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)\Big\}.$$

Why transform? Faster convergence to normality, and the CI respects $0 \le S(t) \le 1$.

2.3 Log-log transformation

Theorem 6.2: Log-log CI

$$\Big(\hat{S}(t)^{1/\theta},\ \hat{S}(t)^{\theta}\Big), \qquad \theta = \exp\Big\{\frac{Z_{1-\alpha/2}\,\sigma_S(t)}{\log \hat{S}(t)}\Big\}.$$

Derivation [Supplement]. K&M use $g(u) = \log\{-\log u\}$, with derivative $g'(u) = 1/(u \log u)$ and inverse $g^{-1}(v) = \exp\{-\exp(v)\}$, so

$$|g'(\hat{S})|\hat{S}\sigma_S = \frac{\sigma_S}{|\log \hat{S}|},$$

and the CI for $g(S)$ is $\log\{-\log \hat{S}\} \pm a$, $a = Z_{1-\alpha/2}\sigma_S/|\log \hat{S}|$. Re-transforming,

$$g^{-1}(\log\{-\log \hat{S}\} \pm a) = \exp\{\log \hat{S} \cdot e^{\pm a}\} = \hat{S}^{e^{\pm a}}.$$

With $\log \hat{S} < 0$, $\theta = e^{-a} < 1$: lower $\hat{S}^{1/\theta}$, upper $\hat{S}^{\theta}$. ■

2.4 Arcsine-square-root transformation

Theorem 6.3: Arcsine-square-root CI

$$\text{lower} = \sin^2\Big\{\max\Big[0,\ \arcsin\big(\hat{S}(t)^{1/2}\big) - 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\},$$

$$\text{upper} = \sin^2\Big\{\min\Big[\frac{\pi}{2},\ \arcsin\big(\hat{S}(t)^{1/2}\big) + 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\}.$$

Derivation [Supplement]. The transformation $g(u) = \arcsin(\sqrt{u})$ has $g'(u) = 1/\{2\sqrt{u(1-u)}\}$, so $|g'(\hat{S})|\hat{S}\sigma_S = 0.5\,\sigma_S(\hat{S}/(1-\hat{S}))^{1/2}$; max / min keep the angle in $[0, \pi/2]$; $g^{-1}(v) = \sin^2 v$. ■

2.5 Log transformation

Definition 6.4: Log CI (default of R `survfit`)

$g(S) = \log S$, $g'(u) = 1/u$, so $|g'(\hat{S})|\hat{S}\sigma_S = \sigma_S$:

$$\min\Big(\exp\{\log \hat{S}(t) \pm Z_{1-\alpha/2}\sigma_S(t)\},\ 1\Big).$$

Less recommended than log-log and arcsine.

Remark 6.5: Comments

- Log-log and arcsine CIs have better coverage than linear; for very large n all agree.
- Transformed CIs are asymmetric, suited to small samples with biased, skewed estimators.

Example 6.6: Toy example, $n = 7$ [Supplement]

T_i	1.5	2.5	1.4	6.2	2.8	5.3	4.5
δ_i	1	0	0	1	1	1	0

Ordered: 1.4^+, 1.5, 2.5^+, 2.8, 4.5^+, 5.3, 6.2.

t_j	d_j	Y_j	$\hat{S}$	σ_S	Linear	Log-log	Arcsine
1.5	1	6	0.833	0.183	(0.535, 1.000*)	(0.273, 0.975)	(0.465, 1.000)
2.8	1	4	0.625	0.342	(0.207, 1.000*)	(0.142, 0.893)	(0.213, 0.949)
5.3	1	2	0.313	0.785	(0.000*, 0.794)	(0.013, 0.734)	(0.006, 0.804)
6.2	1	1	0	—	—	—	—

*Raw linear limits (1.131, 1.043, −0.169) leave $[0,1]$. At $t = 2.8$: $\sigma_S^2 = \frac{1}{30} + \frac{1}{12} = 0.1167$, $\theta = \exp\{1.96 \times 0.3416/\ln 0.625\} = 0.241$, log-log CI $(0.625^{1/0.241}, 0.625^{0.241}) = (0.142, 0.893)$.

3 Pointwise Confidence Intervals for $H(t)$

From $\tilde{H}(t)$ and $\sigma_H(t)$:

- **Linear**: $\tilde{H}(t) \pm Z_{1-\alpha/2}\sigma_H(t)$.
- **Log** (on H; $\mathrm{SE}(\log \tilde{H}) \approx \sigma_H/\tilde{H}$): $\left[\tilde{H}(t)/\phi,\ \phi\tilde{H}(t)\right]$, $\phi = \exp\left[\dfrac{Z_{1-\alpha/2}\sigma_H(t)}{\tilde{H}(t)}\right]$.
- **Arcsine-square-root** (on $S = e^{-H}$):

$$\Big[-2\ln\Big\{\sin\Big(\min\Big[\frac{\pi}{2},\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) + 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\},$$
$$-2\ln\Big\{\sin\Big(\max\Big[0,\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) - 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\}\Big].$$

4 Confidence Bands

4.1 Pointwise intervals vs. bands

A pointwise CI satisfies $P\{L_{t_0} \le S(t_0) \le U_{t_0}\} = 1-\alpha$ for a fixed t_0. Hence a plot of pointwise intervals

- is *not* an interval for the whole curve,
- and is "too narrow" compared with a band.

Definition 6.7: Confidence band

$[L(t), U(t)]$ is a $100(1-\alpha)\%$ confidence band (CB) for $S(t)$ on $t_L \le t \le t_U$ if

$$P\{L(t) \le S(t) \le U(t) \text{ for all } t_L \le t \le t_U\} = 1-\alpha,$$

with t_L, t_U within the range of observed event times.

4.2 Equal probability (EP) bands

- Define

$$a_L = \frac{n\sigma_S^2(t_L)}{1+n\sigma_S^2(t_L)}, \qquad a_U = \frac{n\sigma_S^2(t_U)}{1+n\sigma_S^2(t_U)}, \qquad 0 < a_L \le a_U < 1.$$

- EP bands are **proportional to the pointwise bounds**.

- Replace $Z_{1-\alpha/2}$ in Definition 6.1 and Theorems 6.2–6.3 by $c_\alpha(a_L, a_U)$ from K&M Table C.3 (e.g. $c_{0.10}(0.10, 0.40) = 2.4721 > Z_{0.95} = 1.645$).

4.3 Hall–Wellner (HW) bands

- **Not** proportional to pointwise bounds: wider for small t, narrower for large t.
- With $k_\alpha(a_L, a_U)$ from Table C.4, replace $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 and Theorems 6.2–6.3 by

$$k_\alpha(a_L, a_U)\left[1 + n\sigma_S^2(t)\right] n^{-1/2}.$$

4.4 Performance and sample-size advice

- The EP *linear* band undercovers considerably for $n < 200$.
- Log-log and arcsine EP bands cover about right for $n \geq 20$ (arcsine slightly better).
- All three HW bands do well with as few as 20 events.

EP	Sample size	HW	Sample size
Linear	≥ 200	Linear	≥ 20
Log-log	≥ 20	Log-log	≥ 20
Arcsine	≥ 20	Arcsine	≥ 20

Example 6.8: Bone marrow transplant, ALL patients (K&M Table 4.3)

For 38 ALL patients KM falls from $\hat{S}(1) = 0.9737$ ($Y = 38$) to $\hat{S}(662) = 0.3531$ ($Y = 13$), flat thereafter to 2081 (censored); $\tilde{H}(662) = 1.0152$, $\sigma_H = 0.2185$. On roughly 100–600 days (K&M Fig. 4.5) both bands are wider than the pointwise intervals, and HW is wider than EP early, narrower later.

5 Selected exercises

Motivation. The log CI for H is the log-log CI for S, the arcsine CI for H comes from that for S, and without censoring the band constants reduce to empirical-process objects.

Exercise 6.1 (Log interval for H versus log-log interval for S)

Let $\tilde{H}(t) > 0$ and $\tilde{S}(t) = e^{-\tilde{H}(t)}$.
(a) Show that mapping the log-transformed CI for $H(t)$ through $S = e^{-H}$ gives $(\tilde{S}(t)^\phi, \tilde{S}(t)^{1/\phi})$, which is the log-log CI of Theorem 6.2 with $\hat{S}$ replaced by $\tilde{S}$ and σ_S by σ_H.
(b) Show that $\sigma_H^2(t) \leq \sigma_S^2(t)$ when all $Y_j > d_j$ ($t_j \leq t$), with

$$\sigma_S^2(t) - \sigma_H^2(t) = \sum_{t_j \leq t} \frac{d_j^2}{Y_j^2(Y_j - d_j)}.$$

Solution [Supplement].

(a)

Step 1. $H \mapsto e^{-H}$ is decreasing: $[\tilde{H}/\phi, \phi\tilde{H}] \mapsto [\tilde{S}^{\phi}, \tilde{S}^{1/\phi}]$.

Step 2. With $\hat{S} \to \tilde{S}$, $\sigma_S \to \sigma_H$,

$$\theta = \exp\Big\{\frac{Z_{1-\alpha/2}\sigma_H}{\log \tilde{S}}\Big\} = \exp\Big\{-\frac{Z_{1-\alpha/2}\sigma_H}{\tilde{H}}\Big\} = \frac{1}{\phi}.$$

So $(\tilde{S}^{1/\theta}, \tilde{S}^{\theta}) = (\tilde{S}^{\phi}, \tilde{S}^{1/\phi})$: both are Wald intervals on the $\log(-\log S)$ scale.

(b) Termwise,

$$\frac{d_j}{Y_j(Y_j - d_j)} - \frac{d_j}{Y_j^2} = \frac{d_j\{Y_j - (Y_j - d_j)\}}{Y_j^2(Y_j - d_j)} = \frac{d_j^2}{Y_j^2(Y_j - d_j)} \geq 0;$$

summed over $t_j \leq t$. The relative difference $d_j/(Y_j - d_j)$ matters only for small risk sets (the tail). ■

Exercise 6.2 (Derivation of the arcsine-square-root interval for H)

Derive the arcsine-square-root CI for $H(t)$ stated in this lecture from Theorem 6.3.

Solution [Supplement].

Step 1 (SE on the S scale). $\mathrm{SE}(\tilde{S}) \approx \tilde{S}\,\sigma_H$ (delta method), so apply Theorem 6.3 with $\hat{S} \to \tilde{S}$, $\sigma_S \to \sigma_H$.

Step 2 (rewrite in $\tilde{H}$). $\arcsin(\tilde{S}^{1/2}) = \arcsin(e^{-\tilde{H}/2})$ and

$$\Big(\frac{\tilde{S}}{1-\tilde{S}}\Big)^{1/2} = \Big(\frac{e^{-\tilde{H}}}{1-e^{-\tilde{H}}}\Big)^{1/2} = (e^{\tilde{H}} - 1)^{-1/2}.$$

So the angle interval is $[v_-, v_+]$,

$$v_{\pm} = \mathrm{clip}_{[0,\pi/2]}\Big\{\arcsin\left(e^{-\tilde{H}/2}\right) \pm 0.5 Z_{1-\alpha/2}\sigma_H (e^{\tilde{H}} - 1)^{-1/2}\Big\},$$

and the CI for S is $[\sin^2 v_-, \sin^2 v_+]$.

Step 3 (back to H). Since $H = -2\log\sin v$ decreases in $v \in (0, \pi/2]$, the CI for H is

$$[-2\log\sin v_+,\ -2\log\sin v_-],$$

the stated formula; if $v_- = 0$ the upper limit is $+\infty$. ■

Exercise 6.3 (Band constants without censoring)

Let $a(t) = n\sigma_S^2(t)/\{1 + n\sigma_S^2(t)\}$, so that $a_L = a(t_L)$ and $a_U = a(t_U)$.
(a) Show that a is nondecreasing in t with values in $[0, 1)$.
(b) Without censoring, show that $a(t) = 1 - \hat{S}(t) = \mathbb{F}_n(t)$ whenever $\hat{S}(t) > 0$.
(c) Without censoring, show that the linear HW band is $\hat{S}(t) \pm k_\alpha(a_L, a_U)\, n^{-1/2}$, a band of constant width.

Solution [Supplement].

(a) $\sigma_S^2(t)$ is nondecreasing and $x \mapsto x/(1+x)$ maps $[0, \infty)$ increasingly onto $[0, 1)$.

(b) By Exercise 5.1(b), $\sigma_S^2 = (1 - \hat{S})/(n\hat{S})$, so $1 + n\sigma_S^2 = 1/\hat{S}$ and

$$a(t) = \frac{(1-\hat{S})/\hat{S}}{1/\hat{S}} = 1 - \hat{S}(t) = \mathbb{F}_n(t)$$

by Exercise 5.1(a): $[a_L, a_U]$ is the range of $\mathbb{F}_n$ on $[t_L, t_U]$, the Brownian-bridge time scale.

(c) By (b), $k_\alpha[1 + n\sigma_S^2(t)]n^{-1/2} = k_\alpha n^{-1/2}/\hat{S}(t)$, so the half-width is $k_\alpha n^{-1/2}$: a Kolmogorov–Smirnov-type band. ■

Lecture 7. Estimation of Basic Quantities for Other Sampling Schemes

1 Estimating the Mean Survival Time

1.1 Mean as the area under the curve

Since $\mu = \int_0^\infty S(t)\,\mathrm{d}t$, estimate $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. If the last observation is censored, $\hat{S}$ never reaches 0 and the integral diverges; the estimate then depends on how the tail is completed.

1.2 First solution: Efron's completion

$\hat{S}(t) = 0$ for $t > t_{\max}$ gives $\hat{\mu} = \int_0^{t_{\max}} \hat{S}(t)\,\mathrm{d}t$; it assumes survivors at $t_{\max}$ die immediately (biased downward), and $t_{\max}$ is random.

1.3 Second solution: restricted mean

Theorem 7.1: Restricted mean

For a pre-specified τ, estimate $\mu_\tau = \mathrm{E}[\min(X,\tau)] = \int_0^\tau S(t)\,\mathrm{d}t$ by

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)\,\mathrm{d}t, \qquad \hat{V}(\hat{\mu}_\tau) = \sum_{j=1}^{D} \Big[\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t\Big]^2 \frac{d_j}{Y_j(Y_j - d_j)},$$

with $100(1-\alpha)\%$ CI $\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$ (sum over $t_j \le \tau$).

Remark 7.2: Why the weights $\int_{t_j}^\tau \hat{S}$

Perturbing $\hat{h}_j$ rescales $\hat{S}(t)$ for $t \ge t_j$, changing only the area right of t_j; by the delta method its variance contribution is (that area)^2 times Greenwood's term (Exercise 7.2).

$\hat{\mu}_\tau = \sum_k \hat{S}(s_k)(s_{k+1} - s_k)$ over consecutive jump points s_k in $[0,\tau]$.

Example 7.3: 6-MP group (data of Lecture 5) [Supplement]

$\hat{S} = 0.857, 0.807, 0.753, 0.690, 0.627, 0.538, 0.448$ at $t_j = 6,7,10,13,16,22,23$ ($Y_j = 21,17,15,12,11,7,6$); $\tau = 35$:

$$\begin{aligned}\hat{\mu}_{35} &= 6(1) + 1(0.857) + 3(0.807) + 3(0.753) + 3(0.690)\\ &\quad + 6(0.627) + 1(0.538) + 12(0.448) \approx 23.29 \text{ weeks},\\ \hat{V}(\hat{\mu}_{35}) &\approx 7.99, \quad \mathrm{SE} \approx 2.83, \quad 95\%\ \mathrm{CI} \approx (17.75,\ 28.83).\end{aligned}$$

2 Quantiles and Their Confidence Intervals

$\hat{x}_p = \inf\{t : \hat{S}(t) \le 1-p\}$; its variance involves the hard-to-estimate $f(x_p)$, so CIs **invert** pointwise CIs for $S(t)$.

Theorem 7.4: CIs for x_p

A $100(1-\alpha)\%$ CI for x_p is the set of all t satisfying

$$\text{linear:} \quad -Z_{1-\alpha/2} \le \frac{\hat{S}(t)-(1-p)}{\hat{V}^{1/2}[\hat{S}(t)]} \le Z_{1-\alpha/2},$$

$$\text{log-log:} \quad \left| \frac{[\log\{-\log \hat{S}(t)\} - \log\{-\log(1-p)\}]\hat{S}(t)\log \hat{S}(t)}{\hat{V}^{1/2}[\hat{S}(t)]} \right| \le Z_{1-\alpha/2},$$

$$\text{arcsine:} \quad \left| \frac{2\{\arcsin\sqrt{\hat{S}(t)} - \arcsin\sqrt{1-p}\}\{\hat{S}(t)(1-\hat{S}(t))\}^{1/2}}{\hat{V}^{1/2}[\hat{S}(t)]} \right| \le Z_{1-\alpha/2}.$$

Derivation: where these come from

Each is a Wald test of $H_0 : S(t) = 1-p$ on the transformed scale, $|g(\hat{S}) - g(1-p)|/(|g'(\hat{S})|\,\mathrm{SE}(\hat{S})) \le Z$, with $1/|g'(\hat{S})| = |\hat{S}\log\hat{S}|$ (log-log) or $2\sqrt{\hat{S}(1-\hat{S})}$ (arcsine); collect the non-rejected t.

Example 7.5: Median for the 6-MP group

$\hat{S}(22) = 0.538 > 0.5 \ge \hat{S}(23) = 0.448$, so $\hat{x}_{0.5} = 23$. All three criteria hold from $t = 13$ on ($\hat{S} = 0.690$, SE 0.107), not for $t \le 10$, and never fail again: 95% CI $[13, \infty)$.

3 Right-Censored and Left-Truncated Data

Definition 7.6: Data and modified risk set

Data $\{(L_i, T_i, \delta_i)\}$: entry age L_i, death or censoring time T_i. At death times t_j, d_j deaths and

$$Y_j = \sum_{i=1}^{n} I(L_i \le t_j \le T_i),$$

those *entered* and still observed at t_j.

With this Y_j, KM, Nelson–Aalen, Greenwood, CIs and bands (K&M 4.2–4.4) apply unchanged.

Remark 7.7: What is being estimated

Subjects are seen only after entry, so the targets are conditional:

$$S_L(t) = P(X > t \mid X > L), \qquad H_L(t) = -\ln P(X > t) + \ln P(X > L), \qquad H_L'(t) = h(t).$$

The hazard is unaffected by conditioning, so d_j/Y_j still estimates it (Exercise 7.3).

Example 7.8: Channing House

Channing House residents (97 males, 365 females); truncation times are entry ages (months). The risk set is small at young ages, peaks around 950–1000 months, then falls.

- Males: the first two entered at 751, 759 and died at 777, 781; the third entered at 782.

- Plain KM: factors 1/2 at 777 ($Y = 2$) and 0 at 781 ($Y = 1$), so $\hat{S}_L(t) = 0$ for $t \geq 781$, destroyed by the tiny early risk set.
- Remedy: estimate $S_a(t) = P(X > t \mid X \geq a)$ for an age a with a reasonable risk set:

$$\hat{S}_a(t) = \prod_{a \leq t_j \leq t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}_a(t)] = \hat{S}_a(t)^2 \sum_{a \leq t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)},$$

$$\hat{H}_a(t) = \sum_{a \leq t_j \leq t} \frac{d_j}{Y_j}, \qquad t \geq a.$$

4 Left-Censored Data: Reversing the Time Scale

X failure time, C left censoring time; observe $T = \max(X, C)$, $\delta = I(X \geq C)$.

Theorem 7.9: Reversed time-scale trick

Pick $\tau > \max_i T_i$. Then $\tau - X_i$ is *right* censored by $\tau - C_i$:

$$\tau - T_i = \min(\tau - X_i,\ \tau - C_i), \qquad \delta_i = I(\tau - X_i \leq \tau - C_i).$$

With $G(u) = P(\tau - X > u)$, KM on $\{(\tau - T_i, \delta_i)\}$ gives $\hat{G}$, and

$$\hat{S}(u) = 1 - \hat{G}(\tau - u), \qquad \mathrm{Var}[\hat{S}(u)] = \mathrm{Var}[\hat{G}(\tau - u)].$$

Likewise for CIs and bands.

Why [Supplement]. $\max(X, C) = \tau - \min(\tau - X, \tau - C)$, $X \geq C \iff \tau - X \leq \tau - C$, and $G(\tau - u) = P(X < u)$. ■

5 Right-Truncated Data

We observe $\{(R_i, X_i)\}$ given $X_i < R_i$. With $\tau > \max_i R_i$, $\tau - X_i$ is *left truncated* by $\tau - R_i$; apply the left-truncation method to $\tau - X_i$ given $\tau - X_i > a$.

Definition 7.10: Choice $a = \tau - \tau^*$, $\tau^* = \max_i R_i$

$$G^*(t) = P(\tau - X_i \geq t \mid \tau - X_i \geq \tau - \tau^*) = P(\tau - X_i \geq t \mid X_i \leq \tau^*),$$
$$S^*(t) = P(X_i > t \mid X_i \leq \tau^*) = 1 - G^*(\tau - t).$$

Theorem 7.11: KM on the reversed scale

With $\tau - t_j$ the D distinct failure times on the reversed scale,

$$\hat{G}^*(t) = \prod_{\tau - \tau^* \leq \tau - t_j \leq t} \Big(1 - \frac{d_j}{Y_j}\Big),$$

$$d_j = \#\{i : \tau - X_i = \tau - t_j\}, \quad Y_j = \#\{i : \tau - X_i \geq \tau - t_j \geq \tau - R_i\},$$

and $\hat{S}^*(t) = 1 - \hat{G}^*(\tau - t)$, with inference from the LT properties of $\hat{G}^*$.

Nothing in the data speaks about $X > \tau^*$.

Remark 7.12: Quick review

KM and NA extend to LT + RC data (modified risk set) and to left-censored or right-truncated data (reversed time).

6 Other Sampling Schemes

Interval and double censoring (with or without LT) need Turnbull's iterative methods (Lecture 8).

7 Selected exercises

Motivation. The identity behind μ_τ, its variance by the delta method, and why the modified risk set estimates the hazard.

Exercise 7.1 (The restricted mean)

Let $X \geq 0$ have survival function S and let $\tau > 0$.
(a) Show that $\mathrm{E}[\min(X, \tau)] = \int_0^\tau S(t)\,\mathrm{d}t$.
(b) Without censoring and with $\tau \geq X_{(n)}$, show that $\hat{\mu}_\tau = \bar{X}$.

Solution [Supplement].

(a) $\min(x, \tau) = \int_0^\tau I(t < x)\,\mathrm{d}t$; by Tonelli

$$\mathrm{E}[\min(X, \tau)] = \int_0^\tau P(X > t)\,\mathrm{d}t = \int_0^\tau S(t)\,\mathrm{d}t.$$

(b) By Exercise 5.1(a), $\hat{S}(t) = n^{-1}\sum_i I(X_i > t)$, zero for $t \geq X_{(n)}$:

$$\hat{\mu}_\tau = \frac{1}{n}\sum_{i=1}^n \int_0^\tau I(t < X_i)\,\mathrm{d}t = \frac{1}{n}\sum_{i=1}^n \min(X_i, \tau) = \bar{X}.$$

■

Exercise 7.2 (Variance of the restricted mean)

Derive the variance estimator $\hat{V}(\hat{\mu}_\tau)$ of this lecture by the delta method, under the same approximations as Greenwood's formula (Lecture 5): the $\hat{h}_j = d_j/Y_j$ are approximately uncorrelated with $\mathrm{Var}(\hat{h}_j) \approx \hat{h}_j(1 - \hat{h}_j)/Y_j$; assume $Y_j > d_j$ for $t_j \leq \tau$.

Solution [Supplement].

Step 1. $\hat{S}(t) = \prod_{t_j \leq t}(1 - \hat{h}_j)$, so for $t_j \leq \tau$

$$\frac{\partial \hat{S}(t)}{\partial \hat{h}_j} = -\frac{\hat{S}(t)}{1 - \hat{h}_j} I(t \geq t_j).$$

Step 2. Hence

$$\frac{\partial \hat{\mu}_\tau}{\partial \hat{h}_j} = -\frac{1}{1 - \hat{h}_j}\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t =: -\frac{A_j}{1 - \hat{h}_j}.$$

Step 3. With uncorrelated $\hat{h}_j$,

$$\mathrm{Var}(\hat{\mu}_\tau) \approx \sum_{t_j \le \tau} \frac{A_j^2}{(1-\hat{h}_j)^2} \cdot \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \le \tau} A_j^2 \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \le \tau} A_j^2 \frac{d_j}{Y_j(Y_j - d_j)},$$

which is $\hat{V}(\hat{\mu}_\tau)$; with $\hat{S}(t)$ in place of A_j this is Greenwood's formula. ■

Exercise 7.3 (Why the modified risk set works under left truncation)

Let X be discrete with support $\{x_j\}$ and hazard $h(x_j) = P(X = x_j)/P(X \ge x_j)$. Let the entry time L and censoring time C be such that X, L, C are mutually independent; a subject is sampled only if $L \le X$ and then yields (L, T, δ) with $T = \min(X, C)$, $\delta = I(X \le C)$. Show that, for every x_j with $P(L \le x_j \le T) > 0$,

$$P(T = x_j,\ \delta = 1 \mid L \le x_j \le T,\ \text{sampled}) = h(x_j).$$

Solution [Supplement].

Step 1. $\{L \le x_j \le T\} \subseteq \{L \le X\}$ (sampling), so

$$P(\cdot \mid L \le x_j \le T,\ \text{sampled}) = P(\cdot \mid L \le x_j \le T).$$

Step 2. By independence,

$$\frac{P(X = x_j,\ C \ge x_j,\ L \le x_j)}{P(X \ge x_j,\ C \ge x_j,\ L \le x_j)} = \frac{P(X = x_j)P(C \ge x_j)P(L \le x_j)}{P(X \ge x_j)P(C \ge x_j)P(L \le x_j)} = h(x_j).$$

The numerator is $P(T = x_j, \delta = 1,\ L \le x_j \le T)$, the denominator $P(L \le x_j \le T)$.

So in the modified risk set the chance of an observed death at t_j is $h(t_j)$; multiplying $1 - h$ over $x_j \ge a$ gives only $P(X > t \mid X \ge a)$. ■

Lecture 8. Estimation of Basic Quantities for Other Sampling Schemes (Part 2)

1 Other Sampling Schemes

- Interval censoring: $X_i \in (L_i, R_i]$, never observed exactly.
- Double censoring: each observation is LC, RC, or exact.
- LT + interval censoring, or LT + double censoring.

No simple closed-form estimator; inference is hard.

2 Turnbull's Iterative Method (1974) for Double Censoring

Samples with both left and right censoring are common; Turnbull (1974) gave an iterative modified product-limit estimator.

- **Idea.** Start from S ignoring the left-censored data, compute the *expected* deaths they contribute at each time, update S, repeat (self-consistency / EM).
- **Notation.** Distinct times $0 = t_0 < t_1 < \cdots < t_m$; at t_j: d_j deaths, c_j left-censored, r_j right-censored.

Definition 8.1: Turnbull (1974) algorithm

Step 0. Initial $S_0(t_j)$, e.g. KM ignoring left-censored observations.

Step $(K{+}1).1$. For each left-censoring time t_i and $j \le i$, estimate $p_{ij} = P(t_{j-1} < X \le t_j \mid X \le t_i)$ by

$$p_{ij} = \frac{S_K(t_{j-1}) - S_K(t_j)}{1 - S_K(t_i)}.$$

Step $(K{+}1).2$. Estimated events: $\hat{d}_j = d_j + \sum_{i=j}^{m} c_i\, p_{ij}$.

Step $(K{+}1).3$. KM from the pseudo-data ($\hat{d}_j$ events, r_j right-censored): $Y_j = \sum_{l=j}^{m} (\hat{d}_l + r_l)$.

Repeat until $S_{K+1}(t_j) \approx S_K(t_j)$ for all t_j.

Remark 8.2: Why p_{ij} has this form

Given $X \le t_i$, the event lies in $(t_{j-1}, t_j]$ with probability $[S(t_{j-1}) - S(t_j)]/[1 - S(t_i)]$. Fractional events are the E-step; recomputing KM is the M-step.

Example 8.3: Age at first marijuana use (K&M Section 1.17)

California high-school boys: left censored = used but cannot recall first age; right censored = never used. Initial KM without the 12 left-censored boys (K&M Table 5.1):

t_i	10	11	12	13	14	15	16	17	18	> 18	Total
c_i	0	0	0	1	2	3	2	3	1	0	12
d_i	4	12	19	24	20	13	3	1	0	4	100
r_i	0	0	2	15	24	18	14	6	0	0	79
Y_i	179	175	163	142	103	59	28	11	4	4	
$S_0(t_i)$	.978	.911	.804	.669	.539	.420	.375	.341	.341	0	

Step 1 at $t_4 = 13$ ($1 - S_0(13) = 0.331$):

$$p_{41} = \tfrac{1.000-0.978}{0.331} = 0.067, \quad p_{42} = \tfrac{0.978-0.911}{0.331} = 0.202,$$

$$p_{43} = \tfrac{0.911-0.804}{0.331} = 0.320, \quad p_{44} = \tfrac{0.804-0.669}{0.331} = 0.410.$$

Likewise for all i with $c_i > 0$ (K&M Table 5.2; e.g. column $i = 9$: $0.034, 0.102, 0.161, 0.206, 0.197, 0.180, 0.068, 0.052, 0.000$); Step 2 gives

$$\hat{d}_1 = 4 + 0.067(1) + 0.048(2) + 0.039(3) + 0.036(2) + 0.034(3) + 0.034(1) = 4.487.$$

Step 3 (K&M Table 5.3).

t_i	10	11	12	13	14	15	16	17	18	> 18
$\hat{d}_i$	4.487	13.461	21.313	26.963	22.437	14.714	3.417	1.207	0	4
Y_i	191.000	186.513	173.052	149.739	107.775	61.338	28.624	11.207	4	4
$S_1(t_i)$	.977	.906	.794	.651	.516	.392	.345	.308	.308	0

Iterate to convergence. $Y_1 = 191 = 179 + 12$: the left-censored boys now count as fractional events.

3 Turnbull's Iterative Method (1976) for Interval-Censored Data

Let $0 = \tau_0 < \cdots < \tau_m$ contain all L_i, R_i, and $\alpha_{ij} = I\{(\tau_{j-1}, \tau_j] \subseteq (L_i, R_i]\}$ (subject i could fail at τ_j).

Definition 8.4: Turnbull (1976) algorithm

Step 1. $p_j = S(\tau_{j-1}) - S(\tau_j)$, $j = 1, \ldots, m$.

Step 2. Expected events: $d_j = \sum_{i=1}^{n} \frac{\alpha_{ij} p_j}{\sum_k \alpha_{ik} p_k}$ (each unit mass split over admissible points $\propto p_j$).

Step 3. At risk: $Y_j = \sum_{k=j}^{m} d_k$.

Step 4. Update the product-limit estimate; stop when it stabilizes, else repeat.

Example 8.5: Time to cosmetic deterioration, early breast cancer (radiotherapy only)

46 patients; intervals $(L, R]$ in months or right-censored "$\geq$" values: $(0, 7]$; $(0, 8]$; $(0, 5]$; $(4, 11]$; $(5, 12]$; $(5, 11]$; $(6, 10]$; $(7, 16]$; $(7, 14]$; $(11, 15]$; $(11, 18]$; ≥ 15; ≥ 17; $(17, 25]$; $(17, 25]$; ≥ 18; $(19, 35]$; $(18, 26]$; ≥ 22; ≥ 24; ≥ 24; $(25, 37]$; $(26, 40]$; $(27, 34]$; ≥ 32; ≥ 33; ≥ 34; $(36, 44]$; $(36, 48]$; ≥ 36; ≥ 36; $(37, 44]$; ≥ 37; ≥ 37; ≥ 37; ≥ 38; ≥ 40; ≥ 45; ≥ 46 ($\times 7$).

- **Initial estimate**: spread each mass $1/46$ equally over the grid points in $(L_i, R_i]$, e.g. $(0,7]$ gives $(1/46)(1/4)$ to $\tau = 4,5,6,7$.
- With S_0: $p_1 = 1 - 0.979 = 0.021$, $p_2 = 0.979 - 0.955 = 0.024$, etc.
- $\tau = 4$ lies in $(0,8]$, $(0,7]$, $(0,5]$:

$$d_1 = \frac{0.021}{0.021+0.024+0.021+0.029+0.031} + \frac{0.021}{0.021+0.024+0.021+0.029} + \frac{0.021}{0.021+0.024} = 0.842.$$

- One iteration changes S little early ($0.979 \to 0.982$ at $\tau = 4$), more in the tail ($0.229 \to 0.341$ at $\tau = 46$).

4 The Cohort Life Table

4.1 Setting

A **cohort** shares a time origin. Event or censoring times are recorded only as falling in one of $k+1$ adjacent intervals $(a_{j-1}, a_j]$; the cohort life table summarizes these grouped data.

Example 8.6: NLSY weaning data

NLSY (K&M Section 1.14): 927 breast-fed first-born infants; breast-feeding duration in weeks, weaned (event) or lost (censored). Weeks 0–2: 927 entering, 2 withdrawn; 2–3: 848, 3; 3–5: 774, 6; ...

4.2 Construction

Definition 8.7: Life-table quantities

1. $I_j = (a_{j-1}, a_j]$, $a_0 = 0$, $a_{k+1} = \infty$.
2. Y'_j: number entering I_j event-free.
3. W_j: number withdrawn alive in I_j.
4. $Y_j = Y'_j - W_j/2$: at risk, withdrawals counted for half the interval (*actuarial* adjustment).
5. d_j: number of events in I_j.
6. Survival at the start of I_j ($\hat{S}(a_0) = 1$):

$$\hat{S}(a_j) = \hat{S}(a_{j-1})[1 - d_j/Y_j] = \prod_{i=1}^{j} [1 - d_i/Y_i]. \tag{8.1}$$

7. Density at the midpoint a_{mj}:

$$\hat{f}(a_{mj}) = \frac{\hat{S}(a_{j-1}) - \hat{S}(a_j)}{a_j - a_{j-1}}. \tag{8.2}$$

8. Hazard at the midpoint ($\hat{S}(a_{mj})$ = average of endpoint values):

$$\hat{h}(a_{mj}) = \frac{\hat{f}(a_{mj})}{\hat{S}(a_{mj})} = \frac{2\hat{f}(a_{mj})}{\hat{S}(a_j) + \hat{S}(a_{j-1})} = \frac{d_j}{(a_j - a_{j-1})(Y_j - d_j/2)}. \tag{8.3}$$

Derivation: deriving (8.3)

With $\hat{q}_j = d_j/Y_j$, $\hat{p}_j = 1 - \hat{q}_j$, $\Delta_j = a_j - a_{j-1}$: $\hat{S}(a_j) = \hat{S}(a_{j-1})\hat{p}_j$, $\hat{f}(a_{mj}) = \hat{S}(a_{j-1})\hat{q}_j/\Delta_j$, so

$$\hat{h}(a_{mj}) = \frac{2\hat{S}(a_{j-1})\hat{q}_j/\Delta_j}{\hat{S}(a_{j-1})(1+\hat{p}_j)} = \frac{2\hat{q}_j}{\Delta_j(1+\hat{p}_j)} = \frac{2d_j/Y_j}{\Delta_j(2 - d_j/Y_j)} = \frac{d_j}{\Delta_j(Y_j - d_j/2)}.$$

Events per unit person-time, failures in I_j credited with half the interval.

4.3 Standard errors

Theorem 8.8: Estimated standard deviations

$$\widehat{\mathrm{SD}}[\hat{S}(a_{j-1})] = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{d_i}{Y_i(Y_i - d_i)}} = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i}}, \tag{8.4}$$

$$\widehat{\mathrm{SD}}[\hat{f}(a_{mj})] = \frac{\hat{S}(a_{j-1})\hat{q}_j}{a_j - a_{j-1}}\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i} + \frac{\hat{p}_j}{Y_j\hat{q}_j}}, \tag{8.5}$$

$$\widehat{\mathrm{SD}}[\hat{h}(a_{mj})] = \left\{\frac{1 - [\hat{h}(a_{mj})(a_j - a_{j-1})/2]^2}{Y_j\hat{q}_j}\right\}^{1/2} \hat{h}(a_{mj}). \tag{8.6}$$

(8.4) is Greenwood's formula; $\hat{S}(a_0) = 1$ has SD 0.

4.4 Other summary quantities

1. **Mean**: $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$, completing a flat tail as in Lecture 5.
2. **Median** by interpolation in the I_j with $\hat{S}(a_{j-1}) \geq 0.5 \geq \hat{S}(a_j)$:

$$\hat{x}_{0.5} = a_{j-1} + \frac{[\hat{S}(a_{j-1}) - 0.5](a_j - a_{j-1})}{\hat{S}(a_{j-1}) - \hat{S}(a_j)} = a_{j-1} + \frac{\hat{S}(a_{j-1}) - 0.5}{\hat{f}(a_{mj})}. \tag{8.7}$$

3. **Mean residual life** at the start of I_i:

$$\widehat{\mathrm{mr}}(a_{i-1}) = \frac{\int_{a_{i-1}}^\infty \hat{S}(t)\,\mathrm{d}t}{\hat{S}(a_{i-1})} = \frac{\sum_{j=i}^k \hat{S}(a_{j-1})(a_j - a_{j-1})}{\hat{S}(a_{i-1})}.$$

4. **Median residual life**: $\mathrm{mdrl}(a_{i-1})$ solves $S(a_{i-1})/2 = S\big(a_{i-1} + \mathrm{mdrl}(a_{i-1})\big)$.

Example 8.9: life table for weaning (K&M Table 5.6)

Weeks	Y_j'	W_j	Y_j	d_j	$\hat{S}(a_{j-1})$	$\hat{f}(a_{mj})$	$\hat{h}(a_{mj})$	SD $\hat{S}$	SD $\hat{f}$	SD $\hat{h}$
0–2	927	2	926	77	1.0000	0.0416	0.0434	0	0.0045	0.0049
2–3	848	3	846.5	71	0.9168	0.0769	0.0875	0.0091	0.0088	0.0104
3–5	774	6	771	119	0.8399	0.0648	0.0836	0.0121	0.0055	0.0076
5–7	649	9	644.5	75	0.7103	0.0413	0.0618	0.0149	0.0046	0.0071
7–11	565	7	561.5	109	0.6276	0.0305	0.0537	0.0160	0.0027	0.0051
11–17	449	5	446.5	148	0.5058	0.0279	0.0662	0.0166	0.0021	0.0053
17–25	296	3	294.5	107	0.3381	0.0154	0.0555	0.0158	0.0014	0.0052
25–37	186	0	186	74	0.2153	0.0071	0.0414	0.0138	0.0008	0.0047
37–53	112	0	112	85	0.1296	0.0061	0.0764	0.0114	0.0006	0.0066
53–	27	0	27	27	0.0313	—	—	0.0059	—	—

Row 1: $Y_1 = 926$, $\hat{q}_1 = 0.0832$, $\hat{S}(2) = 0.9168$, $\hat{f} = 0.0416$, $\hat{h} = 77/\{2(926 - 38.5)\} = 0.0434$.
Median: $\hat{S}(11) = 0.5058 \geq 0.5 \geq \hat{S}(17) = 0.3381$, so $\hat{x}_{0.5} = 11 + \dfrac{(0.5058 - 0.5)(17 - 11)}{0.5058 - 0.3381} \approx$ 11.2 weeks.
Hazard (K&M Fig. 5.3): about 0.08–0.09 per week in weeks 0–5, 0.05–0.07 to week 25, lowest (0.04) in 25–37, rising in 37–53.

5 Selected exercises

Motivation. What Turnbull's algorithm converges to, where the actuarial hazard (8.3) comes from, and its SE (8.6).

Exercise 8.1 (Turnbull's algorithm and the likelihood)

In the interval-censored setting, let $p = (p_1, \ldots, p_m)$ range over the simplex $\Delta = \{p_j \geq 0,\ \sum_j p_j = 1\}$, assume each subject has some $\alpha_{ij} = 1$, and let

$$\ell(p) = \sum_{i=1}^{n} \log\Big(\sum_{j=1}^{m} \alpha_{ij} p_j\Big), \qquad D_j(p) = \frac{\partial \ell}{\partial p_j} = \sum_{i=1}^{n} \frac{\alpha_{ij}}{\sum_k \alpha_{ik} p_k},$$

(subject i contributes $P(L_i < X \leq R_i)$), at p with $\ell(p) > -\infty$.
(a) Show that one pass of Steps 1–4 maps p to $T(p)$ with $T(p)_j = d_j/n = p_j D_j(p)/n$.
(b) Show that every maximizer of ℓ over Δ is a fixed point of T.
(c) Conversely, show that a fixed point p of T with $D_j(p) \leq n$ for every j with $p_j = 0$ maximizes ℓ over Δ.

Solution [Supplement].

(a)

Step 1. By Step 2, $d_j = p_j D_j$; each subject contributes total mass 1, so $\sum_j d_j = \sum_j p_j D_j = n$.

Step 2. The pseudo-data are uncensored: $Y_1 = n$, $Y_{j+1} = Y_j - d_j$, so as in Exercise 5.1(a) $S_{\text{new}}(\tau_j) = Y_{j+1}/n$ and $S_{\text{new}}(\tau_{j-1}) - S_{\text{new}}(\tau_j) = d_j/n$.

(b)

Step 1. At a maximizer p all $\sum_k \alpha_{ik} p_k > 0$; for $q \in \Delta$ the directional derivative along $q - p$ is

$$\sum_j D_j(p)(q_j - p_j) = \sum_j D_j q_j - n \leq 0.$$

Step 2. $q = e_j$: $D_j \leq n$ for all j.

Step 3. If $0 < p_j < 1$, $q = (p - p_j e_j)/(1 - p_j) \in \Delta$ gives $p_j(n - D_j)/(1 - p_j) \le 0$, i.e. $D_j \ge n$; if $p_j = 1$, $D_j = n$. So $D_j = n$ where $p_j > 0$.

Step 4. Hence $T(p)_j = p_j D_j / n = p_j$.

(c)

Step 1. $T(p) = p$ gives $D_j = n$ where $p_j > 0$; with the hypothesis, $D_j \le n$ for all j.

Step 2. For $q \in \Delta$ with $\ell(q) > -\infty$, $g(\varepsilon) = \ell(p + \varepsilon(q - p))$ is concave, so $g(1) \le g(0) + g'(0+)$:

$$\ell(q) \le \ell(p) + \sum_j D_j(q_j - p_j) \le \ell(p) + n\sum_j q_j - n = \ell(p).$$

So Turnbull's iteration is an EM scheme for the NPMLE; limits with some $p_j = 0$ must be checked against $D_j \le n$. ■

Exercise 8.2 (The actuarial hazard is a piecewise-exponential MLE)

Suppose the hazard is constant on each life-table interval, $h(x) = h_j$ for $x \in I_j$, and right-censored data (T_i, δ_i) are observed exactly.
(a) Show that the MLE is $\hat{h}_j = d_j/E_j$, where $E_j = \sum_i e_{ij}$ is the total time at risk in I_j, $e_{ij} = \max\{0, \min(T_i, a_j) - a_{j-1}\}$.
(b) If every death and withdrawal in I_j is placed at the midpoint of I_j, show that $\hat{h}_j$ equals (8.3).

Solution [Supplement].

(a)

Step 1. Subject i contributes $h(T_i)^{\delta_i} e^{-H(T_i)}$ (Lecture 4), with $H(T_i) = \sum_j h_j e_{ij}$ and $h(T_i) = h_j$ for a death in I_j.

Step 2. So

$$L(h_1, \dots, h_{k+1}) = \prod_j h_j^{d_j} e^{-h_j E_j}, \qquad \log L = \sum_j \{d_j \log h_j - h_j E_j\}.$$

For $d_j > 0$ the term is strictly concave with maximizer d_j/E_j; for $d_j = 0$ it is maximized at 0.

(b) $Y_j' - d_j - W_j$ survivors contribute Δ_j each and the $d_j + W_j$ others $\Delta_j/2$:

$$E_j = \Delta_j\Big\{Y_j' - d_j - W_j + \frac{d_j + W_j}{2}\Big\} = \Delta_j\Big(Y_j' - \frac{W_j}{2} - \frac{d_j}{2}\Big) = \Delta_j\Big(Y_j - \frac{d_j}{2}\Big),$$

which gives (8.3). ■

Exercise 8.3 (Standard error of the life-table hazard)

Treat $\hat{q}_j = d_j/Y_j$ as a binomial proportion, $\mathrm{Var}(\hat{q}_j) \approx \hat{q}_j\hat{p}_j/Y_j$. Derive (8.6) by the delta method.

Solution [Supplement].

Step 1. $\hat{h} = g(\hat{q})$, $g(q) = \dfrac{2q}{\Delta(2 - q)}$ (index j dropped), and

$$g'(q) = \frac{2(2 - q) + 2q}{\Delta(2 - q)^2} = \frac{4}{\Delta(2 - q)^2}, \qquad \widehat{\mathrm{SD}}(\hat{h}) \approx \frac{4}{\Delta(2 - \hat{q})^2}\Big(\frac{\hat{q}\hat{p}}{Y}\Big)^{1/2}.$$

Step 2. With $u = \hat{h}\Delta/2 = \hat{q}/(2 - \hat{q})$: $1 + u = 2/(2 - \hat{q})$, $1 - u = 2\hat{p}/(2 - \hat{q})$, $1 - u^2 = 4\hat{p}/(2 - \hat{q})^2$.

Step 3. So the right side of (8.6) is

$$\hat{h}\Big(\frac{1-u^2}{Y\hat{q}}\Big)^{1/2} = \frac{2\hat{q}}{\Delta(2-\hat{q})} \cdot \frac{2}{2-\hat{q}}\Big(\frac{\hat{p}}{Y\hat{q}}\Big)^{1/2} = \frac{4}{\Delta(2-\hat{q})^2}\Big(\frac{\hat{q}\hat{p}}{Y}\Big)^{1/2},$$

as in Step 1. ■

Lecture 9. Hypothesis Testing (Part 1)

1 Overview

1.1 One-sample problems

Rationale. A KM pattern may suggest a parametric model (efficient, handles complex sampling), whose fit must then be checked:

$$H_0 : S(t) = S_0(t), \qquad H_0 : H(t) = H_0(t), \qquad H_0 : h(t) = h_0(t) \qquad \text{for all } t \le \tau,$$

with S_0, H_0, h_0 *known*, e.g. $H_0(t) = \lambda t$ for $\text{Exp}(\lambda)$.

1.2 Two- and K-sample problems

For heterogeneous populations, compare groups:

$$H_0 : S_1(t) = S_2(t) \iff H_1(t) = H_2(t) \iff h_1(t) = h_2(t), \qquad t \in [0, \tau],$$

hypotheses of *no* known form; K samples: $H_0 : S_1(t) = \cdots = S_K(t)$.

1.3 How to test?

Without censoring: t or Kolmogorov–Smirnov tests, ANOVA, Wilcoxon or Kruskal–Wallis. The difficulty is censoring.

2 One-Sample Test with Right-Censored Data

Data $\{(T_i, \delta_i)\}$, X_i with hazard h. Test $H_0 : h(t) = h_0(t)$ (equivalently $H(t) = H_0(t)$) on $[0, \tau]$, τ typically the largest study time; t_j, d_j, Y_j as before.

Definition 9.1: One-sample weighted statistic

With a weight $W(t)$, $W(t) = 0$ when $Y(t) = 0$, compare Nelson–Aalen increments with the hypothesized hazard:

$$Z(\tau) = O(\tau) - E(\tau) = \sum_{j=1}^{D} W(t_j) \frac{d_j}{Y(t_j)} - \int_0^\tau W(s) h_0(s)\, \mathrm{d}s.$$

Under H_0

$$V[Z(\tau)] = \int_0^\tau W^2(s) \frac{h_0(s)}{Y(s)}\, \mathrm{d}s,$$

and $Z(\tau)^2 / V[Z(\tau)] \sim \chi^2_1$ approximately.

Remark 9.2: Intuition

Z is a weighted distance $\int W\, \mathrm{d}(\tilde{H} - H_0)$; each increment d_j/Y_j has (Poisson) variance $\approx h_0\, \mathrm{d}s/Y$.

- χ^2: two-sided alternatives only.

- One-sided: reject for large positive $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$ against $h > h_0$, large negative against $h < h_0$.

2.1 One-sample log-rank test

Theorem 9.3: $W(t) = Y(t)$: observed vs. expected

With left truncation (T_i time on study, L_i entry time),

$$O(\tau) = \sum_{j=1}^{D} Y(t_j)\frac{d_j}{Y(t_j)} = \sum_{j=1}^{D}\sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1) = \sum_{i=1}^{n} I(\delta_i = 1),$$

$$E(\tau) = \int_0^\tau Y(s)h_0(s)\,\mathrm{d}s = \sum_{i=1}^{n}\int_0^\tau I(L_i \le s \le T_i)h_0(s)\,\mathrm{d}s = \sum_{i=1}^{n}\left[H_0(T_i) - H_0(L_i)\right].$$

O: observed events by τ; E: expected under H_0; $V[Z(\tau)] = E(\tau)$, so the statistic is $(O - E)^2/E$.

2.2 Harrington–Fleming weights

$$W_{HF}(t) = Y(t)S_0(t)^p[1 - S_0(t)]^q, \qquad p, q \ge 0, \quad S_0(t) = \exp[-H_0(t)].$$

- $p \gg q$: **early** departures emphasized;
- $q \gg p$: **late** departures;
- $p = q$: **mid-range**;
- $p = q = 0$: the one-sample log-rank test.

Example 9.4: IBCSG Trial 6

Multi-center trial of adjuvant chemotherapy, 735 subjects. The NA cumulative hazard is roughly linear up to 100 months (exponential), slope $\hat{\lambda} \approx 0.005$. One-sample log-rank tests ($W = Y$):

H_0	$T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$	p-value
$H_0(t) = 0.0045t$	-0.002	0.998
$H_0(t) = 0.004t$	1.8	0.07
$H_0(t) = 0.005t$	-1.4	0.16
$H_0(t) = 0.006t$	-4.1	< 0.01

$0.0045t$ fits best (a larger hypothesized rate inflates E, making $O - E$ negative).

3 K-Sample Tests

3.1 Hypotheses

$H_0 : h_1(t) = h_2(t) = \cdots = h_K(t) \ \ \forall t \le \tau$ vs. $H_A :$ at least one $h_k(t)$ differs for some $t \le \tau$.

τ is the largest time with all groups at risk, typically $\min_k t_{\max,k}$; the alternative is *global*.

3.2 Data and notation

Data $\{(T_{ik}, \delta_{ik})\}$; $0 < t_1 < \cdots < t_D$ the distinct pooled death times:

- d_j, d_{jk}: deaths at t_j in the pooled sample / in group k;
- Y_j, Y_{jk}: at risk at t_j, pooled / in group k.

3.3 Basic idea and the test statistic

Under H_0 the pooled and within-group NA estimators estimate the same H:

$$\tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j} \to \int_0^t h(u)\,\mathrm{d}u, \qquad \tilde{H}_k(t) = \sum_{t_j \le t} I(Y_{jk} > 0)\frac{d_{jk}}{Y_{jk}} \to \int_0^t h(u)\,\mathrm{d}u.$$

Test a weighted difference of $\tilde{H}_k$ and $\tilde{H}$.

Definition 9.5: Weighted K-sample statistic

With a random weight $W_k(t) \ge 0$, zero when $Y_j = 0$ or $Y_{jk} = 0$:

$$Z_k(\tau) = \sum_{j=1}^{D} W_k(t_j)\Big\{\frac{d_{jk}}{Y_{jk}} - \frac{d_j}{Y_j}\Big\}.$$

Typically $W_k(t_j) = Y_{jk}W(t_j)$ with common W:

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\} \quad = \sum_j W(t_j)\,(\text{observed} - \text{expected deaths in group } k).$$

3.4 Distribution under H_0

Theorem 9.6: Asymptotic normality (via martingale theory)

Under H_0, $(Z_1(\tau), \ldots, Z_K(\tau)) \approx \mathrm{MVN}(\mathbf{0}, \Sigma)$, $\Sigma = (\sigma_{kk'})_{K\times K}$, with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k = 1, \ldots, K,$$

$$\hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}\,d_j, \qquad k \ne k'.$$

Derivation: hypergeometric origin of the variance

Given the margins at t_j, under H_0 the d_j deaths are a random draw from the Y_j at risk, so $(d_{j1}, \ldots, d_{jK})$ is multivariate hypergeometric (Exercise 9.1):

$$\mathrm{E}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}, \quad \mathrm{Var}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}, \quad \mathrm{Cov}(d_{jk}, d_{jk'}) = -d_j\frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}.$$

$(Y_j - d_j)/(Y_j - 1)$ is the finite-population correction (1 if $d_j = 1$). Sum over (approximately uncorrelated) death times.

3.5 Linear dependency and the final statistic

Theorem 9.7: $\sum_k Z_k(\tau) = 0$

$$\sum_{k=1}^{K} Z_k(\tau) = \sum_{j=1}^{D} W(t_j) \sum_{k=1}^{K} \left\{ d_{jk} - d_j \frac{Y_{jk}}{Y_j} \right\} = \sum_{j=1}^{D} W(t_j)(d_j - d_j) = 0,$$

using $\sum_k d_{jk} = d_j$, $\sum_k Y_{jk} = Y_j$: Σ is singular.

Drop one $Z_K(\tau)$:

$$(Z_1, \ldots, Z_{K-1}) \approx \text{MVN}(\mathbf{0}, \Sigma_{(K-1)\times(K-1)}),$$

and

$$\chi^2 = (Z_1(\tau), \ldots, Z_{K-1}(\tau))\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, (Z_1(\tau), \ldots, Z_{K-1}(\tau))^T \overset{H_0}{\sim} \chi^2_{K-1}.$$

Reject if $\chi^2 > \chi^2(K-1, 1-\alpha)$; the value does not depend on the dropped Z_k (Exercise 9.3).

3.6 Special case $K = 2$

$$Z = \frac{Z_1(\tau)}{\sqrt{\hat{V}(Z_1(\tau))}} = \frac{\sum_{j=1}^{D} W(t_j)(d_{j1} - Y_{j1} d_j / Y_j)}{\sqrt{\sum_{j=1}^{D} W^2(t_j) \frac{Y_{j1}}{Y_j}(1 - \frac{Y_{j1}}{Y_j}) \frac{Y_j - d_j}{Y_j - 1} d_j}} \overset{H_0}{\approx} N(0,1).$$

Reject if $|Z| > Z_{1-\alpha/2}$, or use one-sided tests; for $K > 2$ only the two-sided χ^2 is available.

4 Choice of the Weight Function

Weights tune power against specific alternatives; all have $W_k(t_j) = Y_{jk} W(t_j)$.

1. **Log-rank**: $W(t) = 1$; optimal against **proportional hazards** $h_1 = c_2 h_2 = \cdots = c_K h_K$.
2. **Gehan** (generalized Wilcoxon): $W(t) = Y(t)$, so $Z_k(\tau) = \sum_j (Y_j d_{jk} - d_j Y_{jk})$.
3. **Tarone–Ware**: $W(t) = f(Y(t))$, increasing f (original $f(u) = u^{1/2}$; also $\log u$); more weight to large risk sets.
4. **Peto–Peto**: $W(t) = \tilde{S}(t) = \prod_{t_j \le t} (1 - \frac{d_j}{Y_j + 1})$ (pooled).
5. **Modified Peto–Peto** (Andersen, 1982): $W(t) = \tilde{S}(t) Y(t)/(Y(t)+1)$.
6. **Fleming–Harrington**: $W(t_j) = \hat{S}(t_{j-1})^p [1 - \hat{S}(t_{j-1})]^q$, $\hat{S}$ the pooled KM:
   - $p = q = 0$: log-rank;
   - $p > 0, q = 0$: sensitive to early differences;
   - $p = 0, q > 0$: sensitive to late differences.

Remark 9.8: Why the left-continuous $\hat{S}(t_{j-1})$

The weight at t_j must be *predictable* (known just before t_j), as martingale theory requires.

5 Catheter Placement in Kidney Dialysis Patients

Time to exit-site infection for catheters placed surgically (group 1) vs. percutaneously (group 2) (K&M Section 1.4). Surgery is better early, percutaneous later (about 0.8 vs. 0.2 near month 27): the KM curves *cross*, so the weight matters:

- early weighting (e.g. Gehan) may favour surgery;
- late weighting may favour percutaneous placement.

Relative weights (K&M Fig. 7.2): Gehan and Tarone–Ware decrease steeply, log-rank is flat, FH with $q > 0$ increase.

Example 9.9: Log-rank computation (K&M Table 7.2, selected rows)

t_j	Y_{j1}	d_{j1}	Y_{j2}	d_{j2}	Y_j	d_j	$Y_{j1}\frac{d_j}{Y_j}$	$d_{j1} - Y_{j1}\frac{d_j}{Y_j}$	$\frac{Y_{j1}}{Y_j}\left(1-\frac{Y_{j1}}{Y_j}\right)\frac{Y_j-d_j}{Y_j-1}d_j$
0.5	43	0	76	6	119	6	2.168	−2.168	1.326
1.5	43	1	60	0	103	1	0.417	0.583	0.243
2.5	42	0	56	2	98	2	0.857	−0.857	0.485
4.5	36	2	43	0	79	2	0.911	1.089	0.490
							⋮		
26.5	2	1	3	0	5	1	0.400	0.600	0.240
SUM		15		11		26	11.036	3.964	6.211

$Z_1(\tau) = 3.964$, $\hat{\sigma}_{11} = 6.211$, $\chi^2 = 2.53$, $p = 0.112$: not significant.

Example 9.10: Comparison of two-sample tests (K&M Table 7.3)

Test	$W(t_j)$	$Z_1(\tau)$	σ^2_{11}	χ^2	p-value
Log-rank	1	3.96	6.21	2.53	0.112
Gehan	Y_j	−9	38862	0.002	0.964
Tarone–Ware	$Y_j^{1/2}$	13.20	432.83	0.40	0.526
Peto–Peto	$\tilde{S}(t_j)$	2.47	4.36	1.40	0.237
Modified Peto–Peto	$\tilde{S}(t_j)Y_j/(Y_j+1)$	2.31	4.20	1.28	0.259
FH $p=0, q=1$	$1-\hat{S}(t_{j-1})$	1.41	0.21	9.67	0.002
FH $p=1, q=0$	$\hat{S}(t_{j-1})$	2.55	4.69	1.39	0.239
FH $p=1, q=1$	$\hat{S}(t_{j-1})[1-\hat{S}(t_{j-1})]$	1.02	0.11	9.83	0.002
FH $p=0.5, q=0.5$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{0.5}$	2.47	0.66	9.28	0.002
FH $p=0.5, q=2$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^2$	0.32	0.01	8.18	0.004

Remark 9.11: Comments

- Gehan weights early comparisons heavily: *negative* $Z_1(\tau)$.
- FH tests with $q > 0$ are significant: the curves diverge for large t.
- Choose the weight *before* looking at the data.
- Log-rank is most popular, then Gehan.

6 Three-Group Comparison (Bone Marrow Transplant)

Disease-free survival after transplant: 38 ALL, 54 AML low-risk, 45 AML high-risk. Test $H_0 : S_1 = S_2 = S_3$ on $t \le \tau = 2204$ days.

Example 9.12: Log-rank test with $K = 3$

$Z_1(2204) = 2.148$, $Z_2(2204) = -14.966$,

$$\hat{\Sigma} = \begin{pmatrix} 15.9552 & -10.3451 \\ -10.3451 & 20.3398 \end{pmatrix}, \qquad \chi^2 = (2.148,\ -14.966)\,\hat{\Sigma}^{-1} \begin{pmatrix} 2.148 \\ -14.966 \end{pmatrix} = 13.8037.$$

On 2 df, $p \approx 0.001$.

Derivation: evaluating the quadratic form [Supplement]

$\det \hat{\Sigma} = 324.53 - 107.02 = 217.51$ and $\hat{\Sigma}^{-1} = \frac{1}{217.51} \begin{pmatrix} 20.3398 & 10.3451 \\ 10.3451 & 15.9552 \end{pmatrix}$. Then

$$\begin{aligned} \chi^2 &= \frac{1}{217.51}\Big[20.3398(2.148)^2 + 2(10.3451)(2.148)(-14.966) + 15.9552(-14.966)^2\Big] \\ &= \frac{93.85 - 665.14 + 3573.73}{217.51} \approx 13.80. \end{aligned}$$

Gehan $\chi^2 = 16.24$ ($p = 0.0003$); Tarone–Ware 15.65 (0.0004); FH $(1,0)$ 15.67 (0.0004); FH $(0,1)$ 6.11 (0.047); FH $(1,1)$ 9.93 (0.007): all indicate different curves.

7 Selected exercises

Motivation. The hypergeometric moments behind $\hat{\Sigma}$, the log-rank test as a score test, and invariance of χ^2 to the dropped group.

Exercise 9.1 (Hypergeometric moments)

$Y \ge 2$ at risk, Y_k in group k, and the d deaths a uniform random d-subset; d_k the deaths in group k. Prove the stated $\mathrm{E}(d_k)$, $\mathrm{Var}(d_k)$, $\mathrm{Cov}(d_k, d_{k'})$.

Solution [Supplement].

Step 1. With I_i indicating death of subject i: $P(I_i = 1) = d/Y$, $P(I_i = I_l = 1) = \binom{Y-2}{d-2}/\binom{Y}{d} = \dfrac{d(d-1)}{Y(Y-1)}$ $(i \ne l)$.

Step 2. So $\mathrm{Var}(I_i) = \frac{d}{Y}(1 - \frac{d}{Y})$ and

$$\mathrm{Cov}(I_i, I_l) = \frac{d(d-1)}{Y(Y-1)} - \frac{d^2}{Y^2} = \frac{d\{(d-1)Y - d(Y-1)\}}{Y^2(Y-1)} = -\frac{d(Y-d)}{Y^2(Y-1)} =: -c.$$

Step 3. $d_k = \sum_{i \in \text{group } k} I_i$: $\mathrm{E}(d_k) = dY_k/Y$ and

$$\mathrm{Var}(d_k) = Y_k \frac{d(Y-d)}{Y^2} - Y_k(Y_k - 1)c = \frac{Y_k d(Y-d)}{Y^2}\Big\{1 - \frac{Y_k - 1}{Y - 1}\Big\} = d\,\frac{Y_k}{Y}\Big(1 - \frac{Y_k}{Y}\Big)\frac{Y-d}{Y-1}.$$

Step 4. For $k \neq k'$, over $Y_k Y_{k'}$ distinct pairs,

$$\operatorname{Cov}(d_k, d_{k'}) = -Y_k Y_{k'} c = -d\,\frac{Y_k Y_{k'}}{Y^2}\,\frac{Y-d}{Y-1}.$$

■

Exercise 9.2 (The one-sample log-rank test is a score test)

Let $T = \min(X, C)$, $\delta = I(X \le C)$ with $X \perp\!\!\!\perp C$, X continuous with cumulative hazard H.
(a) Show that $\mathrm{E}(\delta) = \mathrm{E}\{H(T)\}$.
(b) In the exponential model $h \equiv \lambda$ (no truncation), show that the score statistic for $H_0 : \lambda = \lambda_0$, with the information estimated by $\sum_i T_i/\lambda_0$, equals the one-sample log-rank statistic $(O-E)^2/E$.

Solution [Supplement].

(a) With $G(s) = P(C \ge s)$, $H(T) = \int_0^\infty I(T \ge s) h(s)\,\mathrm{d}s$, Tonelli and independence,

$$\mathrm{E}\{H(T)\} = \int_0^\infty P(X \ge s) G(s) h(s)\,\mathrm{d}s = \int_0^\infty f(s) G(s)\,\mathrm{d}s = P(X \le C) = \mathrm{E}(\delta),$$

as $P(X \ge s)h(s) = f(s)$. (The counting process minus its compensator has mean zero.)

(b)

Step 1. With $O = \sum_i \delta_i$, $\Sigma T = \sum_i T_i$: $\ell = O \log\lambda - \lambda\,\Sigma T$, $U(\lambda) = O/\lambda - \Sigma T$, $\mathcal{I}(\lambda) = \mathrm{E}(O)/\lambda^2$.

Step 2. By (a), $\mathrm{E}(O) = \lambda_0 \mathrm{E}(\Sigma T)$ under H_0, so $\mathcal{I}(\lambda_0)$ is estimated by $\Sigma T/\lambda_0$.

Step 3. $E = \sum_i H_0(T_i) = \lambda_0 \Sigma T$ ($L_i = 0$), so

$$\frac{U(\lambda_0)^2}{\Sigma T/\lambda_0} = \frac{(O - \lambda_0\Sigma T)^2/\lambda_0^2}{\Sigma T/\lambda_0} = \frac{(O-E)^2}{\lambda_0 \Sigma T} = \frac{(O-E)^2}{E}.$$

■

Exercise 9.3 (The χ^2 does not depend on the dropped group)

Let $\mathbf{Z} = (Z_1, \dots, Z_K)'$ and $\hat\Sigma$ be as in this lecture. For each k let $\mathbf{Z}_{(-k)}$ and $\hat\Sigma_{(-k)}$ be obtained by deleting the k-th entry, respectively the k-th row and column, and assume $\hat\Sigma_{(-K)}$ is invertible.
(a) Show that $\hat\Sigma \mathbf{1} = \mathbf{0}$.
(b) Show that $\hat\Sigma_{(-k)}$ is invertible and $\mathbf{Z}'_{(-k)} \hat\Sigma^{-1}_{(-k)} \mathbf{Z}_{(-k)} = \mathbf{Z}'_{(-K)} \hat\Sigma^{-1}_{(-K)} \mathbf{Z}_{(-K)}$ for every k.

Solution [Supplement].

(a) At a death time with $\pi_k = Y_{jk}/Y_j$, $w = W(t_j)^2 d_j (Y_j - d_j)/(Y_j - 1)$, the contribution to $\sum_{k'} \hat\sigma_{kk'}$ is

$$w\Big\{\pi_k(1-\pi_k) - \sum_{k' \neq k} \pi_k \pi_{k'}\Big\} = w\,\pi_k\Big\{1 - \sum_{k'} \pi_{k'}\Big\} = 0.$$

(b)

Step 1. It suffices to compare $k = 1$ with K. With P_k deleting coordinate k, $\mathbf{Z}_{(-k)} = P_k \mathbf{Z}$, $\hat\Sigma_{(-k)} = P_k \hat\Sigma P_k'$.

Step 2. On $V = \{\mathbf{z} : \mathbf{1}'\mathbf{z} = 0\}$, $P_1 \mathbf{z} = M P_K \mathbf{z}$ with $M : (z_1, \dots, z_{K-1}) \mapsto (z_2, \dots, z_{K-1}, -\sum_{k<K} z_k)$, invertible.

Step 3. $\mathbf{Z} \in V$, and by (a) and symmetry every column of $\hat{\Sigma}$ lies in V. Applying Step 2 to the columns of $\hat{\Sigma}P_1'$, then of $\hat{\Sigma}P_K'$,

$$P_1\hat{\Sigma}P_1' = MP_K\hat{\Sigma}P_1' = M\big(P_1\hat{\Sigma}P_K'\big)' = M\big(MP_K\hat{\Sigma}P_K'\big)' = M\hat{\Sigma}_{(-K)}M'.$$

Step 4. So $\hat{\Sigma}_{(-1)} = M\hat{\Sigma}_{(-K)}M'$ is invertible, and with $\mathbf{Z}_{(-1)} = M\mathbf{Z}_{(-K)}$,

$$\mathbf{Z}_{(-1)}'\hat{\Sigma}_{(-1)}^{-1}\mathbf{Z}_{(-1)} = \mathbf{Z}_{(-K)}'M'(M')^{-1}\hat{\Sigma}_{(-K)}^{-1}M^{-1}M\mathbf{Z}_{(-K)} = \mathbf{Z}_{(-K)}'\hat{\Sigma}_{(-K)}^{-1}\mathbf{Z}_{(-K)}.$$

■

Lecture 10. Hypothesis Testing (Part 2)

1 Review: The K-Sample Test

Right-censored data $\{(T_{ik}, \delta_{ik})\}$; pooled death times $t_1 < \cdots < t_D$; d_j, d_{jk} deaths and Y_j, Y_{jk} at risk at t_j (pooled / group k). With a common weight W,

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\}, \qquad \chi^2 = (Z_1, \ldots, Z_{K-1})\,\hat{\Sigma}_{K-1}^{-1}\,(Z_1, \ldots, Z_{K-1})^T \overset{H_0}{\approx} \chi^2_{K-1},$$

with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}d_j, \qquad \hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

This omnibus test is **not sensitive to trends**.

2 Tests for Trend

2.1 Order-specific alternatives

$$H_0 : h_1(t) = \cdots = h_K(t) \;\; \forall t \le \tau \qquad \text{vs.} \qquad H_A : h_1(t) \le h_2(t) \le \cdots \le h_K(t) \;\; \forall t \le \tau,$$

with at least one strict inequality ($S_1 \ge \cdots \ge S_K$).

2.2 Trend test statistic

Objective: more power against the ordering.

Definition 10.1: Trend test

For scores $a_1 < a_2 < \cdots < a_K$,

$$Z = \frac{\sum_{k=1}^{K} a_k Z_k(\tau)}{\Big(\sum_{k=1}^{K}\sum_{k'=1}^{K} a_k a_{k'} \hat{\sigma}_{kk'}\Big)^{1/2}} = \frac{\mathbf{a}'\mathbf{Z}}{\sqrt{\mathbf{a}'\hat{\Sigma}\mathbf{a}}},$$

with the full $K \times K$ $\hat{\Sigma}$. Under H_0, $Z \approx N(0,1)$ (a combination of mean-zero martingales); reject if Z exceeds the upper α quantile (one-sided).

- **Choice of** a_k: under H_A, $\mathrm{E}\{Z_1(\tau)\} \le \cdots \le \mathrm{E}\{Z_K(\tau)\}$ (higher hazard, more observed than expected deaths), so the standard $a_k = k$ weights the high-expectation groups; other choices are possible.
- $\hat{\Sigma}$ is singular, but $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ for non-constant $\mathbf{a}$ (Exercise 10.1).

2.3 Invariance

Theorem 10.2: Invariance under linear transformations of the scores

Let $Z_{\mathbf{a}} = \mathbf{a}'\mathbf{Z}/\sqrt{V[\mathbf{a}'\mathbf{Z}]}$ and $\mathbf{b} = c\mathbf{a} + d\mathbf{1}_K$ with constants $c > 0$, d. Then $Z_{\mathbf{b}}(\tau) = Z_{\mathbf{a}}(\tau)$.

Proof [Supplement]. Since $\mathbf{1}'\mathbf{Z} = 0$ (so $\hat{\Sigma}\mathbf{1} = \mathbf{0}$),

$$\mathbf{b}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z} + d\,\mathbf{1}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z}, \qquad V[\mathbf{b}'\mathbf{Z}] = V[c\,\mathbf{a}'\mathbf{Z}] = c^2 V[\mathbf{a}'\mathbf{Z}].$$

Hence $Z_{\mathbf{b}} = Z_{\mathbf{a}}$ (the sign flips if $c < 0$). ■

E.g. $(1, 2, 3, 4)$, $(0, 1, 2, 3)$, $(-3, -1, 1, 3)$ give the same test.

Example 10.3: larynx cancer by stage

90 patients in stages I–IV; the KM curves are ordered (K&M Fig. 7.3). With $a_k = k$ and log-rank weights,

$$\mathbf{Z}(10.7) = (-7.5660,\ -3.0117,\ 2.9155,\ 7.6623),$$

$$\hat{\Sigma} = \begin{pmatrix} 12.0740 & -4.4516 & -6.2465 & -1.3759 \\ -4.4516 & 7.8730 & -2.7599 & -0.6614 \\ -6.2465 & -2.7599 & 9.9302 & -0.9238 \\ -1.3759 & -0.6614 & -0.9238 & 2.9612 \end{pmatrix}.$$

$\mathbf{a}'\mathbf{Z} = 25.806$ and $\mathbf{a}'\hat{\Sigma}\mathbf{a} = 48.152$, so

$$Z = \frac{25.806}{\sqrt{48.152}} = 3.72, \qquad Z^2 = 13.83, \qquad p < 0.0001.$$

(K&M report $Z^2 = 13.83$.) Tarone–Ware $Z = 4.06$, Gehan 4.22, Peto–Peto 4.13; all $p < 0.0001$: the hazards are in the expected order.

Remark 10.4

Use a trend test only with strong prior knowledge of the ordering.

3 Stratified Tests

3.1 Rationale

Comparisons often need covariate adjustment, e.g. 3 vs. 6 months of chemotherapy (IBCSG) adjusting for stage. Regression adjustment is sensitive to misspecification; **stratification** is a conservative alternative.

3.2 Set-up

Definition 10.5: Stratified K-sample test

$H_0 : h_{1s}(t) = h_{2s}(t) = \cdots = h_{Ks}(t)$, $t \in [0, \tau]$, for each stratum $s = 1, \ldots, M$.

1. Within stratum s, compute $Z_{ks}(\tau)$ and $\hat{\Sigma}_s$.
2. Sum: $Z_{k\cdot}(\tau) = \sum_s Z_{ks}(\tau)$, $\hat{\Sigma}_\cdot = \sum_s \hat{\Sigma}_s$ (independent strata).

3. $\chi^2 = [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]^T \overset{H_0}{\sim} \chi^2_{K-1}$.

Observed and expected deaths are compared within strata, so baseline differences across strata cannot masquerade as group effects.

Example 10.6: bone marrow transplant, stratified by MTX

The three disease groups (Lecture 9), now stratified by methotrexate prophylaxis (MTX) or not (NOMTX); log-rank ($W = 1$):

$$\text{MTX:}\quad (Z_1, Z_2, Z_3) = (1.2816,\ -2.4929,\ 1.2113),$$
$$\hat{\Sigma}_{MTX} = \begin{pmatrix} 6.23336 & -3.42915 & -2.80422 \\ -3.42915 & 5.60732 & -2.17818 \\ -2.80422 & -2.17818 & 4.98239 \end{pmatrix},$$
$$\text{NOMTX:}\quad (Z_1, Z_2, Z_3) = (-0.504,\ -11.235,\ 11.739),$$
$$\hat{\Sigma}_{NOMTX} = \begin{pmatrix} 9.6771 & -6.5535 & -3.1236 \\ -6.5535 & 13.9816 & -7.4281 \\ -3.1236 & -7.4281 & 10.5517 \end{pmatrix}.$$

Summing: $Z_{1\cdot} = 0.7776$, $Z_{2\cdot} = -13.7279$, $Z_{3\cdot} = 12.9503$, and

$$\hat{\Sigma}_{\cdot} = \begin{pmatrix} 15.91046 & -9.98265 & -5.92782 \\ -9.98265 & 19.58892 & -9.60628 \\ -5.92782 & -9.60628 & 15.53409 \end{pmatrix}.$$

With the first two:

$$\chi^2 = (0.7776,\ -13.7279) \begin{pmatrix} 15.9105 & -9.9827 \\ -9.9827 & 19.5889 \end{pmatrix}^{-1} \begin{pmatrix} 0.7776 \\ -13.7279 \end{pmatrix} = 13.19 \quad (2 \text{ df},\ p \approx 0.001).$$

Remark 10.7: Remarks

Good power when effects point the *same direction* in every stratum; otherwise they can cancel, and separate tests per stratum are preferable.

4 Matched Pairs

Each matched pair is a stratum: times (T_{1i}, T_{2i}), indicators $(\delta_{1i}, \delta_{2i})$, $H_0 : h_{1i}(t) = h_{2i}(t)$. With $K = 2$, per pair

$$Z_{1i}(\tau) = \sum_j W(t_j)\Big\{d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big\}, \qquad \sigma_{11i} = \sum_j W(t_j)^2 \frac{Y_{j1}}{Y_j}\Big(1 - \frac{Y_{j1}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j.$$

Derivation: within a single pair

Until the first time in the pair, $Y_j = 2$, $Y_{j1} = 1$.

1. $T_{1i} = T_{2i}$: $d_j = \delta_{1i} + \delta_{2i}$,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i} + \delta_{2i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i} - \delta_{2i}}{2},$$

$$\sigma_{11i} = W^2(T_{1i})\tfrac{1}{2}(1-\tfrac{1}{2})\frac{2-\delta_{1i}-\delta_{2i}}{2-1}(\delta_{1i}+\delta_{2i})$$
$$= \frac{W^2(T_{1i})}{4}(2-\delta_{1i}-\delta_{2i})(\delta_{1i}+\delta_{2i}).$$

2. $T_{1i} < T_{2i}$: only the first time counts,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i}}{2}, \qquad \sigma_{11i} = \frac{W^2(T_{1i})}{4}(2-\delta_{1i})\delta_{1i}.$$

3. $T_{1i} > T_{2i}$: $Z_{1i} = -W(T_{2i})\dfrac{\delta_{2i}}{2}, \quad \sigma_{11i} = \dfrac{W^2(T_{2i})}{4}(2-\delta_{2i})\delta_{2i}.$

(Afterwards $Y_{j1}/Y_j \in \{0,1\}$ and terms vanish.) With $(2-\delta)\delta = \delta$:

$$Z_{1i} = \begin{cases} W(T_{1i})/2, & T_{1i} < T_{2i}, \delta_{1i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{1i} = 1, \delta_{2i} = 0, \\ -W(T_{2i})/2, & T_{2i} < T_{1i}, \delta_{2i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{2i} = 1, \delta_{1i} = 0, \\ 0, & \text{otherwise;} \end{cases}$$

$$\hat{\sigma}_{11i} = \begin{cases} W(T_{1i})^2/4, & \text{first case,} \\ W(T_{2i})^2/4, & \text{second case,} \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 10.8: Paired log-rank (sign-type) test

For constant W (e.g. log-rank), with D_1 (D_2) the pairs in which member 1 (2) has the event first,

$$Z_{1\cdot} = \frac{W}{2}(D_1 - D_2), \qquad \hat{\sigma}_{11\cdot} = \sum_{i=1}^{M}\hat{\sigma}_{11i} = \frac{W^2}{4}(D_1+D_2), \qquad \frac{Z_{1\cdot}}{\sqrt{\hat{\sigma}_{11\cdot}}} = \frac{D_1 - D_2}{\sqrt{D_1+D_2}} \overset{H_0}{\approx} N(0,1).$$

A sign (McNemar-type) test: informative pairs go either way with probability $\frac{1}{2}$ under H_0 (Exercise 10.2).

Example 10.9: 6-MP vs. placebo matched pairs

$D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$: $Z = 15/\sqrt{21} = 3.27$, $p = 0.001$.

5 Renyi-Type Tests

5.1 Rationale

If the hazards **cross**, early and late differences cancel and the usual tests lose power. Renyi tests use the *supremum of absolute cumulative differences*, a censored-data Kolmogorov–Smirnov statistic.

5.2 Set-up and statistic

$H_0 : h_1 = h_2$ on $t \le \tau$ vs. $H_A : h_1(t) \neq h_2(t)$ for some t; notation as before with $K = 2$.

Definition 10.10: Renyi statistic

Cumulative weighted difference:

$$Z(t_i) = \sum_{t_j \le t_i} W(t_j)\Big[d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big], \qquad i = 1, \dots, D,$$

with (τ the largest t_j with $Y_{j1}, Y_{j2} > 0$)

$$\sigma^2(\tau) = \sum_{t_j \le \tau} W^2(t_j)\frac{Y_{j1}}{Y_j}\frac{Y_{j2}}{Y_j}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

and two-sided statistic

$$Q = \frac{\sup\{|Z(t)|,\ t \le \tau\}}{\sigma(\tau)}.$$

Theorem 10.11: Null distribution

Under H_0, $Q \approx \sup\{|B(x)|,\ 0 \le x \le 1\}$, B standard Brownian motion (K&M Table C.5).

Remark 10.12: Why Brownian motion

$Z(t)$ is a martingale with variance $\sigma^2(t)$; in time $x = \sigma^2(t)/\sigma^2(\tau)$, $Z/\sigma(\tau)$ behaves like $B(x)$, and the supremum catches excursions even if $Z(\tau) \approx 0$.

Example 10.13: gastric cancer: chemotherapy vs. chemotherapy + radiotherapy

KM curves cross (K&M Fig. 7.5): chemo + radiation worse early, better later. Log-rank Renyi: $|Z(t_j)|$ peaks at $t_j = 315$ with 9.8 (K&M Fig. 7.4); $\sigma(2363) = 4.46$:

$$Q = 9.8/4.46 = 2.2, \qquad 0.05 < p < 0.06 \text{ (K\&M Table C.5)},$$

nearly significant, whereas the log-rank statistic $-2.15/4.46 = -0.482$ gives $p = 0.63$.

6 Selected exercises

Motivation. When the trend variance $\mathbf{a}'\hat{\Sigma}\mathbf{a}$ is positive, and why the paired log-rank test is an exact sign test.

Exercise 10.1 (When is the trend test defined?)

Write $\pi_{jk} = Y_{jk}/Y_j$ and $w_j = W(t_j)^2 d_j(Y_j - d_j)/(Y_j - 1)$, with $w_j = 0$ if $Y_j = 1$.
(a) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} = \sum_j w_j\Big\{\sum_k \pi_{jk}a_k^2 - \Big(\sum_k \pi_{jk}a_k\Big)^2\Big\} \ge 0$.
(b) Show that $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ if and only if at some death time with $w_j > 0$ the scores a_k are not all equal over the groups with $Y_{jk} > 0$. In particular, for $a_1 < \cdots < a_K$ it suffices that at one death time with $W(t_j) \ne 0$ and $d_j < Y_j$ at least two groups are at risk.
(c) For $K = 2$ and $a_1 < a_2$, show that the trend statistic equals $Z_2(\tau)/\sqrt{\hat{\sigma}_{22}} = -Z_1(\tau)/\sqrt{\hat{\sigma}_{11}}$.

Solution [Supplement].

(a) Each t_j contributes $w_j\{\mathrm{diag}(\boldsymbol{\pi}_j) - \boldsymbol{\pi}_j\boldsymbol{\pi}_j'\}$ to $\hat{\Sigma}$; the brace in the formula is the variance of a_κ for a label $\kappa \sim \boldsymbol{\pi}_j$, hence ≥ 0, and $w_j \geq 0$.

(b) The variance vanishes iff a_k is constant on $\{k : Y_{jk} > 0\}$, and a sum of nonnegative terms is positive iff some term is. For increasing scores, two groups at risk make the a_k unequal, and $W(t_j) \neq 0$, $1 \leq d_j < Y_j$ give $w_j > 0$.

(c) $Z_1 = -Z_2$ and $\hat{\sigma}_{11} = -\hat{\sigma}_{12} = \hat{\sigma}_{22}$ (Exercise 9.3(a)), so

$$\mathbf{a}'\mathbf{Z} = (a_2 - a_1)Z_2, \qquad \mathbf{a}'\hat{\Sigma}\mathbf{a} = (a_2 - a_1)^2\hat{\sigma}_{22}.$$

Divide $(a_2 - a_1 > 0)$: the one-sided two-sample test of $h_1 \leq h_2$. ■

Exercise 10.2 (The paired log-rank test as an exact sign test)

In pair i ($i = 1, \ldots, M$, pairs independent) let X_{1i}, X_{2i} be i.i.d. continuous and both censored at a common time C_i independent of (X_{1i}, X_{2i}). Let D_1 (D_2) be the number of pairs in which member 1 (member 2) has an observed event first, and $m = D_1 + D_2$.
(a) Show that, conditionally on m, $D_1 \sim \mathrm{Bin}(m, 1/2)$.
(b) Show that $\mathrm{E}(D_1 - D_2 \mid m) = 0$ and $\mathrm{Var}(D_1 - D_2 \mid m) = m$, matching the paired log-rank statistic.
(c) Compute the exact two-sided p-value for the 6-MP pairs ($D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$).

Solution [Supplement].

(a)

Step 1. Member 1 has an observed event first iff $A_{1i} = \{X_{1i} < X_{2i},\ X_{1i} \leq C_i\}$; likewise A_{2i}; disjoint (ties null).

Step 2. Swapping X_{1i}, X_{2i} preserves the law and maps A_{1i} to A_{2i}: equal probabilities.

Step 3. With $\xi_i = 1, 2, 0$ for A_{1i}, A_{2i}, neither: the ξ_i are independent and $P(\xi_i = 1 \mid \xi_i \neq 0) = 1/2$, so given $J = \{i : \xi_i \neq 0\}$, $D_1 \sim \mathrm{Bin}(|J|, 1/2)$, depending on J only through $m = |J|$.

(b) $D_1 - D_2 = 2D_1 - m$: mean 0, variance m; with $W \equiv 1$ this is the paired statistic $(D_1 - D_2)/\sqrt{D_1 + D_2}$.

(c) With $m = 21$ and $D_1 = 18$,

$$p = 2P\{\mathrm{Bin}(21, \tfrac{1}{2}) \geq 18\} = 2 \cdot \frac{\binom{21}{18} + \binom{21}{19} + \binom{21}{20} + \binom{21}{21}}{2^{21}} = \frac{2(1330 + 210 + 21 + 1)}{2\,097\,152} \approx 0.0015,$$

close to the normal approximation 0.001. ■